Chemistry

TARGET : JEE (Main + Advanced)

ENTHUSIAST + LEADER COURSE

GRP # 01 to 12 (BOOKLET)

(PHYSICAL)

(ENGLISH)

Rankers Academy JEE

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GUIDED REVISION
CHEMISTRY(PHYSICAL)

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Important Notes

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SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have (+3, 0) marks.

1. 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C. The concentration of H^+ at equivalence point is
($K_w = 1 \times 10^{-14}$ at 25°C)

(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

2. How many moles of NH_3 must be added to 2.0 litre of 0.80 M $AgNO_3$ in order to reduce the Ag^+ concentration to 5×10^{-8} M. K_f of $[Ag(NH_3)_2^+] = 10^8$

(A) 0.4 (B) 2 (C) 3.52 (D) 4

3. Let solubility of a sparingly soluble salt of strong base and weak acid in pure water is equal to s_1, in a buffer solution with pH < 7 is equal to s_2 and in another buffer solution with pH > 7 is equal to s_3. If hydrolysis is considered in all the solutions then the correct relation between s_1, s_2 and s_3 is

(A) $s_2 = s_2 = s_3$ (B) $s_1 > s_2 > s_3$ (C) $s_1 < s_2 < s_3$ (D) $s_2 > s_1 > s_3$

4. For the reaction : $LiCl.3NH_3(s) \rightleftharpoons LiCl.NH_3(s) + 2NH_3(g)$; $Kp = 9$ atm^2 at 300 K.
In 8.2 L vessel 1.4 mol of $LiCl.NH_3$ (s) and 2.8 moles of NH_3 are placed. Moles of $LiCl.3NH_3$(s) formed at equilibrium will be :

(A) 1.4 (B) 0.7 (C) 0.9 (D) 1

5. Two moles of an equimolar mixture of two alcohols R_1-OH and R_2-OH are esterified with one mole of acetic acid. If only 80% of the acid is consumed till equilibrium and the quantities of ester formed under equilibrium are in the ratio 3 : 2. What is the value of equilibrium constant for the esterification of R_1-OH?

(A) 0.48 (B) 3.69 (C) 2.2 (D) 0.75

6. An amount of 1 mole each of A and D is introduced in 1L container. Simultaneously the following two equilibria are established :

$A \rightleftharpoons B + C$; $K_C = 10^6$M

$B + D \rightleftharpoons A$; $K_C = 10^{-6}$M^{-1}

The equilibrium concentration of A will be

(A) 10^{-6} M (B) 10^{-3} M (C) 10^{-12} M (D) 10^{-4} M

7. Assuming no change in volume, calculate the minimum mass of NaCl necessary to dissolve 0.010 mol AgCl in 100 L solution. [$K_f(AgCl_2^-) = 3 \times 10^5$, $K_{sp} = (AgCl) = 1 \times 10^{-10}$]

(A) $\frac{10}{3}$gm (B) 195 gm (C) 19.5 kg (D) 1950 gm

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8. The degree of hydrolysis of a salt of weak acid and weak base in it's 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be

(A) 100% (B) 50% (C) 25% (D) none of these

SECTION-I (ii) Multiple Correct : (Maximum Marks: 20)

Each question of this section have (+4, 0) marks.

9. Which of the following is true for alkaline aqueous solution?

(A) $pH > \frac{pk_w}{2}$ (B) $pH > pOH$ (C) $pOH < \frac{pk_w}{2}$ (D) $pH < pOH$

10. An amount of 1 mole of $PCl_3(g)$ and 1 mole of $PCl_5(g)$ is taken in a vessel of 10 L capacity maintained at 400 K. At equilibrium, the moles of Cl_2 is found to be 0.004

(A) K_C for the reaction ; $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 0.0004 M

(B) K_p for the reaction ; $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$ is $0.0004 \times (0.0821 \times 400)$ atm.

(C) If $PCl_3(g)$ is added to the equilibrium mixture, ΔG at the new equlibrium becomes greater than the ΔG at old equilibrium

(D) After equilibrium is achieved, moles of PCl_3 are doubled and moles of Cl_2 are halved simultaneously, then the partial pressure of PCl_5 remains unchanged.

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11. At 90°C pure water has $[H^+] = 10^{-6}$ M. Based on this correct statement is (are) -

(A) Increasing the temperature, increases degree of ionization of water.

(B) At 60°C, pH of pure water will be less than 6

(C) At 90°C, pH of a 10^{-6} M HCl solution will be less than 6

(D) Ionization constant (K_a) of H_2O at 90°C is 10^{-12}

12. Select the correct statement(s) :

(A) If pOH of asolution at 70°C is 7, then solution will be acidic

(B) If $[H^+]$ concentration is decreases 100 times, pH of solution increases by 2 units

(C) If solution of CH_3COONa(a) is diluted then pH increases.

(D) Buffer capacity remains constant with dilution.

13. Aqueous solutions of HNO_3, KOH, CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) –

(A) HNO_3 and CH_3COOH (B) KOH and CH_3COONa

(C) HNO_3 and CH_3COONa (D) CH_3COOH and CH_3COONa

SECTION-II Second Decimal : (Maximum Marks: 12)

Each question of this section have (+3, 0) marks.

1. For the reaction : $A(g) \rightleftharpoons 2B(g)$, the equilibrium partial pressures of gases A and B is 4 bar and 2 bar, respectively. If the equilibrium mixture is slowly & isothermally compressed to 12 bar, maintaining equilibrium, then the new equilibrium pressure(in bar) of B(g) becomes.

2. A weak base BOH was titrated against a strong acid . The pH at 1/4th equivalence point was 8.0. Enough strong base was now added (7.5 millimole) to completely convert the salt. The total volume was 10 ml. Find the pOH at this point.

3. How much water (in L) should be added to 10 g CH_3COOH to give pOH equal to 11.
$K_a(CH_3COOH) = 1.8 \times 10^{-5}$

4. If $[H^+]$ of a 1 litre solution made by mixing 0.01 mol HCl with 1 mol BCl is xM, then the value of '1000x' is.
[Given : pK_b of BOH = 10, $\sqrt{5} = 2.236$]

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1.

	Column-I		Column-II
(A)	$2AB_{3(g)} \rightleftharpoons A_{2(g)} + 3B_{2(g)}$	(P)	Introduction of $B_{2(g)}$ at constant volume shift equilibrium in backward direction
(B)	$AB_{3(g)} \rightleftharpoons AB_{(g)} + B_{2(g)}$	(Q)	α increases on increasing temperature
(C)	$2AB_{(g)} \rightleftharpoons A_{2(g)} + B_{2(g)}$	(R)	On adding inert gas at constant pressure equilibrium concentration will change
(D)	$AB_{2(s)} \rightleftharpoons A_{(g)} + B_{2(g)}$	(S)	On increasing the pressure, concentration of all reactants increases
		(T)	α is independent of equilibrium pressure.

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SECTION-I (i) Single Correct (Maximum Marks:27)

Each question of this section have (+3, 0) marks.

1. A buffer solution was prepared by dissolving 0.02 mole acetic acid & 0.01 mole sodium acetate in enough water to make 1.0 L of solution $\left(K_{a(CH_3COOH)} = 2 \times 10^{-5}\right)$. If 5×10^{-4} mole NaOH were added to 100 ml of the buffer. The resultant pH of the solution would be [given log 5 = 0.70]

(A) 5 (B) 4.7 (C) 5.3 (D) None

2. PCl_5 (molecular mass = M) dissociates into PCl_3 and Cl_2 as : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. If the total pressure of the system at equilibrium is P and the density is 'd' at temeprature , T K. The degree of dissociation of PCl_5 may be represented as

(A) $\frac{PM}{dRT}$ (B) $\frac{PM}{dRT} - 1$ (C) $\frac{dRT}{PM} - 1$ (D) $\frac{dRT}{PM}$

3. Solid Ammonium carbamate dissociates as: $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.

(A) $\frac{31}{9}$ (B) $\frac{31}{27}$ (C) $\frac{27}{31}$ (D) $\frac{9}{31}$

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4. Which of the following can not act as buffer solution -

(A) 30 ml 0.1M CH_3COOH + 10 ml 0.3M NH_4OH

(B) 10 ml 0.2M Na_2CO_3 + 20 ml 0.2M H_2CO_3

(C) 30 ml 0.1M HCl + 10 ml 0.1M Na_3PO_4

(D) 10 ml 0.1M NH_4Cl + 5 ml 0.1M NaOH

5. A certain gas 'A' polymerizes to a very small extent at a given temperature as : $nA(g) \rightleftharpoons A_n(g)$. The reaction is started with one mole of 'A' in a container of capacity V. Which of the following is correct value of $\frac{PV}{RT}$, at equilibrium?

(A) $1 - \frac{(n-1).K_C}{V^{n-1}}$ (B) $\frac{(n-1).K_C}{V^{n-1}}$ (C) $1 - \frac{n.K_C}{V^{n-1}}$ (D) $1 - \frac{n.K_C}{V^{n-1}}$

6. The increasing order of Ag^+ ion concentration in :

(I) Saturated solution of AgCl (II) Saturated solution of AgI

(III) 1M $Ag(NH_3)_2^+$ in 0.1 M NH_3 (IV) 1M $Ag(CN)_2^-$ in 0.1 M KCN

Given : $K_{sp}(AgCl) = 10^{-10}$; $K_{sp}(AgI) = 10^{-16}$, $K_f(Ag(NH_3)_2^+) = 10^8$; $K_f(Ag(CN)_2^-) = 10^{11}$

(A) II < I < III < IV (B) IV < III < II < I (C) IV < II < III < I (D) IV < II < I < III

7. pH of a solution by mixing 50 mL of 0.1 M Na_3PO_4 and 50 mL of 0.05 M NaH_2PO_4 is :
[For H_3PO_4 : $pK_1 = 2.1$, $pK_2 = 7.2$, $pK_3 = 12$ and log 3 = 0.3]

(A) 9.6 (B) 4.65 (C) 7.2 (D) 11.7

8. For the reaction
$[Ag(CN)_2]^- \rightleftharpoons Ag^+ + 2CN^-$; K_c at 25°C is 4×10^{-19}
The equilibrium concentration of Ag^+ is, in a solution which is 0.1 M in KCN & 0.03 M in $AgNO_3$ originally.

(A) 7.5×10^{-16}M (B) 7.5×10^{-18} M (C) 1.25×10^{19}M (D) 1.25×10^{-17}M

9. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{th}$ of that of a strong acid (HX, 1M), at 25°C. The K_a of HA is

(A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

Each question of this section have (+4, 0) marks.

10. A reaction at 300K with $\Delta G° = -1743$ J consists of 3 moles of A(g), 6 moles of B(g) and 3 moles of C(g). If A, B and C are in equilibrium in 1 L container, then the reaction may be.
(ln 2 = 0.7, R = 8.3 J/K-mol).

(A) $A + B \rightleftharpoons C$ (B) $A \rightleftharpoons B + 2C$
(C) $2A \rightleftharpoons B + C$ (D) $A + B \rightleftharpoons 2C$

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11. Which of the following combination of solutes would result in the formation of a buffer solution ?

(A) $NH_3 + NH_4Cl$

(B) CH_3COOH + NaOH in 2 : 1 molar ratio

(C) NH_3 + HCl in 2 : 1 molar ration

(D) CH_3COOH + NaOH in 1 : 2 molar ratio

12. Choose the correct alternatives

(A) K_{sp} of $Fe(OH)_3$ in aqueous solution is 3.8×10^{-38} at 298 K. The concentration of Fe^{3+} will increases with decrease in H^+ ion concentration.

(B) In a mixture of NH_4Cl and NH_4OH in water, a further amount of NH_4Cl is added. The pH of mixture will decrease.

(C) An aqueous solution of each of the following salts NH_4I, $Cr(NO_3)_3$, KCN will be basic, acidic and neutral respectively.

(D) Degree of hydrolysis of 0.1 M CH_3COONH_4, 0.2 M CH_3COONH_4, 0.5 M CH_3COONH_4 is same.

SECTION-I (iii) Paragraph : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 13 - 15

	Column-I Compound		Column-II Relationship		Column-III Statement
(A)	CH_3COOH ($K_a = 2 \times 10^{-5}$)	(P)	$pH = \frac{1}{2}(P^{Ka_1} + P^{Ka_2})$	(I)	Addition of HCl(aq) can make buffer solution
(B)	0.1 M $NaHCO_3$	(Q)	$pH = \frac{1}{2}(pK_a - \log C)$	(II)	Increases solubility of AgCl(s)
(C)	NH_3(aq.) ($K_b = 2 \times 10^{-5}$)	(R)	$pH = 7 + \frac{1}{2}(pK_b + \log C)$	(III)	pH is independent of concentration
(D)	CH_3COONa(aq.)	(S)	$pOH = \frac{1}{2}(pK_b - \log C)$	(IV)	Addition of NaOH (aq.) can make buffer solution

13. Which of the following option is incorrectly matched -

(A) A ⟶ Q ⟶ IV

(B) B ⟶ S ⟶ IV

(C) C ⟶ S ⟶ II

(D) C ⟶ S ⟶ I

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14. 30 ml of 0.1 M CH_3COOH (aq.) is mixed with 10 ml of 0.1 M NaOH(aq.) pH of resulting solution

(A) 4.7

(B) 5.3

(C) 4.4

(D) 5.6

15. For H_2CO_3 ($K_{a1} = 10^{-7}$; $K_{a1} = 10^{-11}$)
Number of OH^- in 1 litre of 0.1 M $NaHCO_3$(aq.) is -

(A) $10^{-5} \times N_A$

(B) $10^{-9} \times N_A$

(C) $10^{-7} \times N_A$

(D) $10^{-11} \times N_A$

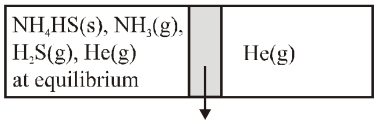
SECTION-II Second Decimal : (Maximum Marks: 12)

Each question of this section have (+3, 0) marks.

1. 10 ml of 0.1M weak acid HA($k_a = 10^{-5}$) is mixed with 10 ml 0.2M HCl and 10 ml 0.1M NaOH. If the value of $[A^-]$ in the resulting solution is $x \times 10^{-5}$ M the value of 'x' is.

2. The solubility of $Pb(OH)_2$ in water is 2.0×10^{-6}M. If the solubility of $Pb(OH)_2$ in a buffer solution of pH = 8 is $x \times 10^{-5}$ M the value of 'x' is.
If your answer is $x \times 10^{-y}$ then fill 'y' in OMR.

3.



Fixed SPM, which allows only He(g) to cross it
The entire system is at equilibrium at 300 K. The volume of each chamber is 82.1 L. The total pressure in left chamber is 4 atm and in right chamber is 2 atm. $NH_3(g)$ and $H_2S(g)$ are obtained only from the dissociation of $NH_4HS(s)$. The value of K_P (in atm^2) for the reaction: $NH_4HS\,(s) \rightleftharpoons NH_3\,(g) + H_2S\,(g)$, is

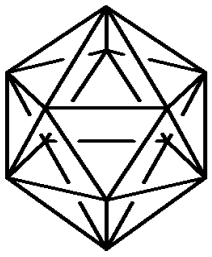
4. $COF_2(g)$ passed over catalyst at 1000°C comes to equilibrium: $2COF_2\,(g) \rightleftharpoons CO_2\,(g) + CF_4\,(g)$. Analysis of the equilibrium mixture (after quick cooling to freeze the equilibrium) shows that 500 ml of the equilibrium mixture (STP) contains 300 ml (STP) of (COF_2 + CO_2) taking the total pressure to be 10 atm. The value of K_P for the reaction is

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SECTION-I (i) Single Correct (Maximum Marks:18)

Each question of this section have (+3, 0) marks.

1. Boron exist in different allotropic forms. All allotropic form contains icosahedral units (icosahedral is a regular shape with 12 corners & 20 faces) with boron atoms at all 12 corners and all bonds are equivalent.



Icosahedral

Calculate $|\Delta H|$ (in kJ) per mole of boron atoms forming gaseous icosahedral if ΔH_{BE} (B –B)= 200 kJ/mol.

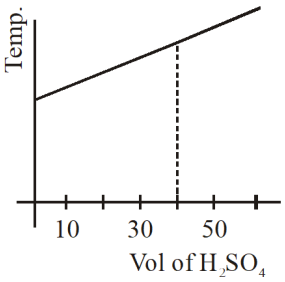
(A) 1000 (B) 600

(C) 500 (D) 250

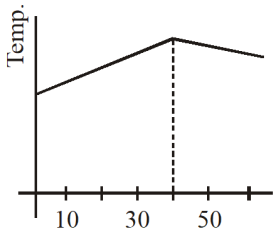
2. In an experiment the enthalpy of neutralisation of sodium hydroxide with sulphuric acid, $50cm^3$ of 0.4M sodium hydroxide were titrated thermometrically with 0.25M sulphuric acid. Which of the following plots give the most probable correct representation temperature of solution (initial temperature of NaOH & H_2SO_4 are same).

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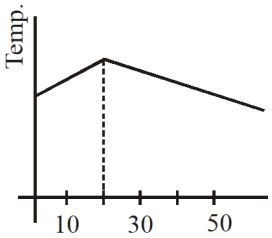
(A)



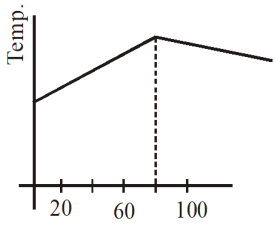
(B)



(C)



(D)



3. 10 litre of an ideal gas at 25 atm & 27°C is expanded isothermally to 1atm against a constant external pressure of 760 torr. Calculate work done by the gas in litre-atm

(A) 240 (B) 250 (C) 800 (D) 400

4. Which of the following reactions is spontaneous only at relatively low temperature -

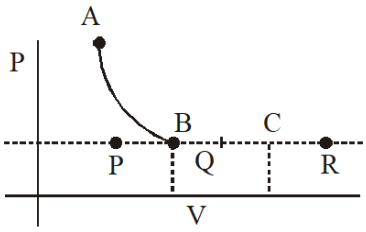
(A) $NH_4Br(s) + 188\ kJ \longrightarrow NH_3(g) + Br_2(\ell)$

(B) $NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s) + 176kJ$

(C) $2H_2O_2(\ell) \longrightarrow 2H_2O(\ell) + O_2(g) + 196\ kJ$

(D) Both (B) & (C)

5. For an ideal gas, three adiabatic processes are carried out upto same final pressure from same initial state. If adiabatic reversible process ends up at point 'B' and adiabatic single step irrversible process ends up at point 'C', then adibatic free expansion upto same final pressure will end up at -



(A) A

(B) P

(C) Q

(D) R

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6. Which of the options contain correct set of parameters which are both amount and path independent parameters.

(I) Internal energy

(II) Specific enthalpy

(III) Heat exchange at constant pressure condition

(IV) Molar heat exchange at constant volume condition

(V) Temperature

(VI) Work involved

(A) Only II, IV & V

(B) All option except VI

(C) All options except III, IV & VI

(D) Only II & V

SECTION-I (ii) Multiple Correct : (Maximum Marks: 20)

Each question of this section have (+4, 0) marks.

7. Assume ideal gas behaviour for all the gases considered & vibrational degrees of freedom to be active. Separate equimolar samples of Ne, O_2, SO_2 & CH_4 were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.

Step-I $\longrightarrow$ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step-II $\longrightarrow$ After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s)

(A) Due to step I only, the decrease in temperature will be maximum for Ne.

(B) During step II, heat given will be minimum for CH_4

(C) There will be no net change in internal energy for any of the gas after both the steps of process are completed.

(D) The P–V graph of CH_4 & SO_2 will be same for overall process.

8. Choose the correct statement(s)

(A) During adiabatic expansion of an ideal gas magnitude of work obtained is equal to ΔH of gas.

(B) For same change in temperature of ideal gas through adiabatic process, magnitude of W will be same in reversible as well as irreversible process

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(C) During boiling of H_2O at 1 atm, 100°C kinetic energy of H_2O molecules increases.

(D) For same change in volume of ideal gas, ΔU is more in adiabatic expansion than in isobaric expansion if expansion starts with same intial state.

9. Select the correct options(s):

(A) ΔH_f^o [$H_2O(\ell)$] is zero at 25°C

(B) ΔS_f^o [$O_2(g)$] is zero at 25°C

(C) S°[$O_2(g)$] is zero at 25°C

(D) Entropies of aqueous ions can be negative or positive

10. Select the incorrect statement (s) -

(A) $\Delta H_{atomisation}$ of graphite is equal to ΔH_f [C(g)]

(B) ΔH_{comb} of 'H' atom is equal to ΔH_f [$H_2O(\ell)$]

(C) ΔH_f [$H_2O(\ell)$] is zero

(D) ΔH_{comb} of graphite is equal ΔH_f [CO(g)]

11. Which of the following is correct option(s)-

($\in$: Bond enthalpy, $\Delta_a H$: enthalpy of atomisation)

(A) $(\Delta_f H)_{H(g)} = \frac{1}{2}(\Delta_a H)_{H_2(g)} = \frac{1}{2}\in_{H-H}$

(B) $(\Delta_f H)_{C(g)} = (\Delta_a H)_{C(gr,s)} = (\Delta_{sub} H)_{C(gr,s)}$

(C) $(\Delta_f H)_{Br(g)} = \frac{1}{2}(\Delta_a H)_{Br_2(\ell)} = \frac{1}{2}\in_{Br-Br}$

(D) $(\Delta_f H)_{I(g)} = (\Delta_a H)_{I_2(s)} = (\Delta_{sub}\ H)_{I_2(s)}$

SECTION-I (iii) Paragraph : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 12 - 14

Study the following thermochemical equation :

(a) $N_2O(g) + 3H_2(g) \longrightarrow H_2O(\ell) + N_2H_4(\ell)$; $\Delta H = -76$ kcal

(b) $4N_2(g) + 3H_2O(\ell) \longrightarrow 2NH_3(g) + 3N_2O(g)$; $\Delta H = +240$ kcal

(c) $N_2H_4(\ell) + H_2O(\ell) \longrightarrow 2NH_3(g) + \frac{1}{2} O_2(g)$; $\Delta H = +36$ kcal

(d) $H_2O(\ell) \longrightarrow H_2(g) + \frac{1}{2} O_2(g)$; $\Delta H = 68$ kcal

12. The enthalpy of combustion (in kcal/mol) of liquid N_2H_4 is -

(A) -129

(B) -138

(C) -147

(D) 147

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13. The enthalpy of formation (in kcal / mol) of liquid N_2H_4 is -

(A) -11

(B) $+44$

(C) $+11$

(D) $+13$

14. Ammonia can be obntained according to reaction (b) or (c). Which of the following is correct regarding these reactions, for the same mass of ammonia formed.

(A) More mass of reactants is needed for reaction (c)

(B) Less mass of reactants is needed for reaction (b)

(C) More heat is needed for reaction (b)

(D) More heat is needed for reaction (c)

SECTION-II Second Decimal : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

1. Pressure over 1000 mL of a liquid is gradually increases from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 mL, and ΔH of the process, assuming linear variation of volume with pressure, is 'x' kJ, then the value of (x) is.

2. One mole of ideal gas undergoes following cyclic process:

(I) Isochoric heating from (P_1 V_1 T_1) to double temperature.

(II) Isobaric expansion from to double volume.

(III) Linear expansion (on PV curve) to (P_1, $8V_1$).

(IV) Isobaric compression to initial state.

If the initial temperature of the gas is 300 K and the magnitude of work done (in kcal) by the gas.

Given: $R = 2\ \frac{Cal}{mol - K}$

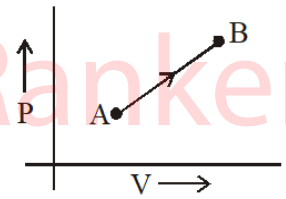
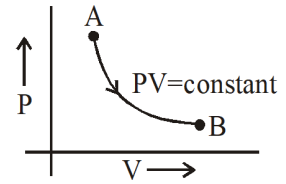
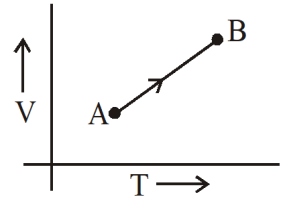
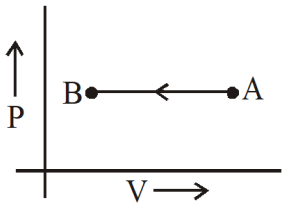
3. $2A(s) \rightleftharpoons B(g) + 2C(g) + 3D(g)$

Total pressure developed in closed container by decomposition of A at equilibrium is 12 bar at 727°C. If $\Delta G°$ (kcal) , of the reaction at 727°C is (R = 2cal/mole-K, ln 2 = 0.7, ln3 = 1.1)

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. One mole of an ideal gas undergoing from state A to State-B through different processes column-I shows graph of the process & column-II shows change in parameter in these process. Match the following

	Column-I		Column-II
(A)		(P)	q > 0
(B)		(Q)	W > 0
(C)		(R)	$\Delta U > 0$
(D)		(S)	$\Delta H > 0$
		(T)	W < 0

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SECTION-I (i) Single Correct (Maximum Marks:27)

Each question of this section have (+3, 0) marks.

1. Each of the vessel 1 and 2 contain 1.2 moles of gaseous helium. The ratio of the vessel volumes is $V_2/V_1 = 2.0$, and the ratio of the absolute temperature of helium in them is $T_1/T_2 = 2.0$. Assuming the gas to be ideal, find the difference of gas entropies in these vessels, $S_2 - S_1$. (ln 2 = 0.7).

(A) 0.84 cal/ K (B) 4.2 cal/ K (C) –0.84 cal/ K (D) – 4.2 cal/ K

2. The entropy change accompanying the transfer of 12000 J of heat from a body A at 327°C to a body B at 127°C is -

(A) –10.0 J/K (B) +10.0 J/K (C) –57.8 J/K (D) +57.8 J/K

3. $2NO_2(g) \longrightarrow N_2O_4(g)$; $\Delta U°_{reaction} = -16$ kcal /mol

If $\Delta U°_f [N_2O_4(g)] = 2$kcal/ mole

then calculate $\Delta H°_{formation}$ of NO_2 (g). All data is at 727°C.

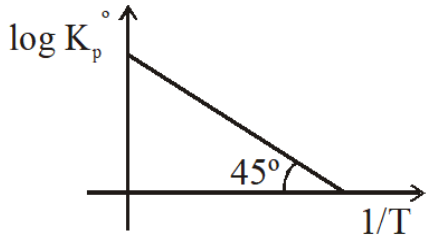
(A) 9 kcal /mol (B) 4.5 kcal /mol (C) 8 kcal /mol (D) 10 kcal /mol

4. For a process which follows the equation $PV^3 = C$, the work done when one mole of ideal gas was taken from 1 atm to $2\sqrt{2}$ atm starting from initial temperature of 300 K is

(A) 300 R (B) 150 R (C) 600 R (D) 900 R

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5. For a reaction graph of log $K_p°$ and 1/T as follows -



then standard enthalpy change for reaction will be -

(A) 2 cal/mol (B) –2 cal/mol (C) 4.606 cal/mol (D) –4.606 cal/mol

6. Which one of the following statement is false.

(A) For a fix amount of a substance entropy of a substance in the liquid phase will be lower than the entropy of the same substance in gaseous phase

(B) Spontaneous process always occur very rapidly

(C) for a fix amount of a solid ΔS sublimation of a solid always positive

(D) If $\Delta G < 0$, then it is not necessary that process will be always spontaneous

7. Heat liberated by a given amount of an ideal gas undergoing reversible isothermal process is 1200 cal at 300K. What is the Gibbs free energy change of the gas in this process ?

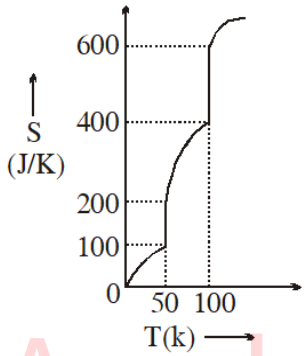
(A) zero (B) +1200 cal (C) −1200 cal (D) 4 cal

8. ΔH_f^0 of water is − 285.8 kJ mol^{-1}. If enthalpy of neutralisation of monoacid strong base is −57.3 kJ mol^{-1}, ΔH_f^0 of OH^- (aq.) ion will be

(A) − 228.5 kJ mol^{-1} (B) 228.5 kJ mol^{-1}

(C) 114.25 kJ mol^{-1} (D) −114.25 kJ mol^{-1}

9. For 1 mole substance variation of absolute entropy with temperature is given by following graph.
What is molar enthalpy of vapourisation (ΔH_{vap}).



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(A) 2 kJ

(B) 5 kJ

(C) 20 kJ

(D) 10 kJ

SECTION-I (ii) Multiple Correct : (Maximum Marks: 12)

Each question of this section have (+4, 0) marks.

10. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes
(A) isothermal expansion (B) adiabatic expansion
(C) free expansion in insulated condition (D) expansion at constant pressure.
If E_1,E_2,E_3 and E_4 respectively are the changes in average kinetic energy of the molecules for the above four processes, then-

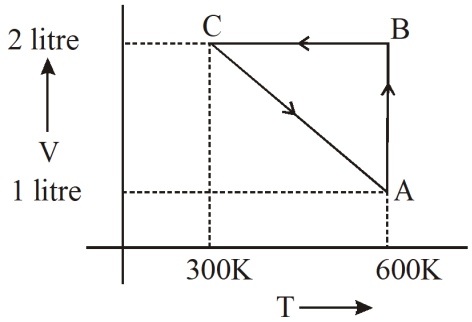
(A) $E_2 = E_3$

(B) $E_1 = E_3$

(C) $E_1 > E_4$

(D) $E_4 > E_3$

11. 1 mole $N_2(g)$ undergo following cyclic process -



which option (s) is/are correct - [Use : R = 2 Cal / K - mole ; ln 2 = 0.7]

(A) $W_{AB} = -840$ cal

(B) $\Delta H_{AB} = 300$ cal

(C) $\Delta U_{CA} = 1500$ cal

(D) $q_{BC} = -600$ cal.

12. For $MX_2(s)$ lattice enthalpy is 800kJ mol^{-1} and enthalpy of solution is 50kJ mol^{-1}. If hydration energies of M^{2+} and X^- ions are in the ratio 3 : 1 , then correct options are :

(A) Hydration energy of M^{2+} is -300 kJ/mol

(B) Hydration energy of M^{2+} is -450 kJ/mol

(C) Hydration energy of X^- is -300 kJ/mol

(D) Hydration energy of X^- is -150 kJ/mol

SECTION-I (iii) Paragraph : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

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Paragraph for Questions 13 - 15

	Column-I (Reactions)		Column-II (Characteristics)		Column-III (Characteristics)
(I)	$C(s) + O_2(g) \longrightarrow CO_2(g)$	(i)	ΔH = +ve	(P)	$\Delta H > \Delta E$
(II)	$SO_3(g) \longrightarrow SO_2(g) + \frac{1}{2}O_2(g)$	(ii)	Δn_g = +ve	(Q)	$\lvert\Delta H\rvert > \lvert\Delta E\rvert$
(III)	$2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$	(iii)	Δn_g = −ve	(R)	$\Delta H < \Delta E$
(IV)	$C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$	(iv)	ΔH = −ve	(S)	$\lvert\Delta H\rvert < \lvert\Delta E\rvert$

13. Which of the following is only correct match ?

(A) I , i , P (B) II , ii , Q (C) III , iii , S (D) IV , iv , R

14. Which of the following is only incorrect match ?

(A) II , i , P (B) III , iv , R (C) IV , iii , S (D) II , ii , Q

15. Which of the following is only correct match ?

(A) I , ii , Q (B) II , iii , Q (C) III , iv , S (D) IV , ii , P

SECTION-II Second Decimal : (Maximum Marks: 12)

Each question of this section have (+3, 0) marks.

1. At 500 kilobar pressure, density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value $|\Delta H - \Delta U|$ (kJ/mole) for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T' :

2. A solution of 8.4 gm of a newly discovered compound (molar mass = 80000) in 25 ml of solution shows a temperature rise of 0.04°C for complete oxygenation. Each mole of compound binds 10 moles of oxygen. If heat capacity of solution is 4.2 J/K- ml then calculate amount of heat released per mole of oxygen bound in kJ

3. The increase in Gibbs free energy (in kJ) of 13 g of ethanol (density = 0.78 g cm^3), when the pressure is increased isothermally from 1 bar to 3001 bar, is

4. The entropy change for the isothermal reversible expansion of an ideal gas at 300K from a particular state A to a state B is 20 J/K. Calculate the work (in kJ) obtained in the process.

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. Match the column

	Column–I		Column–II
(A)	Vaporisation of water at 100°C & 1 atm	(P)	$\Delta S > 0$
(B)	Isothermal reversible expansion of an ideal gas	(Q)	$\Delta H = 0$
(C)	Adiabatic free expansion of ideal gas	(R)	$\Delta G < 0$
(D)	Isobaric heating of an ideal gas	(S)	$W = 0$
		(T)	$q > 0$

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SECTION-I (i) Single Correct (Maximum Marks:27)

Each question of this section have (+3, 0) marks.

1. 10 ml of 0.10 M solution of a compound ($Na_2CO_3.NaHCO_3.2H_2O$) is titrated against 0.05 M HCl. X ml of HCl is used when phenolphthalein is the indicator and Y ml of HCl is used when methyl orange is the indicator in two separate titration, hence (Y – X) is

(A) 20 ml (B) 40 ml

(C) 60 ml (D) None of these

2. Calculate the mass of anhydrous oxalic acid, which can be oxidised to CO_2 (g) by 100 ml of an MnO^-_4 solution, 10 ml of which is capable of oxidising 50 ml of $1NI^-$ to I_2.

(A) 45 gm (B) 22.5 gm

(C) 30 gm (D) 12.25 gm

3. An aqueous solution contains 2M-$AlCl_3$ and 3m-$SnCl_4$. If the molarity and molality of Cl^- ion in the solution is same, the density of solution is (Al = 27, Sn = 119)

(A) 1.05 gm/ ml (B) 1.15 gm/ ml

(C) 1.95 gm / ml (D) 2.05 gm/ ml

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4. A 150 mL of solution of I_2 is divided into two unequal parts. I part reacts with hypo solution in acidic medium. 15 mL of 0.4 M hypo was consumed. II part was added with 100 mL of 0.3 M NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. What was the initial concentration of I_2 ?

(A) 0.08 M (B) 0.1 M

(C) 0.2 M (D) none of these

5. A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+}. In the presence of fluoride ion, Mn^{2+} is titrated with 3 lit of $KMnO_4$ solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III) , What was the % of Mn_3O_4 in the sample?

(A) 50 % (B) 75%

(C) 25% (D) 40%

6. 20 ml of a solution of H_2O_2 is diluted to 100 ml and 50 ml of this solution is titrated with 0.02 M $KMnO_4$ solution in acidic medium and required 50 ml. The volume strength of original H_2O_2 solution is -

(A) 0.285 V (B) 1.12 V

(C) 2.80 V (D) 2.28 V

7. A mixture of $NaHC_2O_4$ and $KHC_2O_4 \cdot H_2C_2O_4$ required equal volumes of 0.2 N $KMnO_4$ and 0.12 N NaOH separately. What is the molar ratio of $NaHC_2O_4$ and $KHC_2O_4 \cdot H_2C_2O_4$ in the mixture ?

(A) 6 : 1 (B) 3 : 1

(C) 1 : 3 (D) none

8. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{M}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{M}{10}$ sodium hydroxide for complete neutralizaton. The percentage of nitrogen in the compound is :

(A) 3% (B) 5%

(C) 6% (D) 10%

9. A mixture of FeO and Fe_2O_3 is reacted with acidified $KMnO_4$ solution having a concentration of 2/5 M, 100 mL of which was used. The solution was then titrated with Zn dust which converted Fe^{3+} of the solution to Fe^{2+}. The Fe^{2+} required 1000 mL of 2/15 M $K_2Cr_2O_7$ solution. Find the ratio by mol of FeO and Fe_2O_3.

(A) 2/3 (B) 3/2

(C) 1/2 (D) 3/1

SECTION-I (ii) Multiple Correct : (Maximum Marks: 8)

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Each question of this section have (+4, 0) marks.

10. A definite volume of pure ammonia (NH_3) gas is passed through a series of electric sparks by which the volume becomes 90 ml. The increase in volume is due to formation of nitrogen (N_2) and hydrogen (H_2) gases. All the gases finally present are washed with dilute sulphuric acid solution, by which the volume of gases becomes 80 ml. All the volume are measured at the same temperature and pressure. Which of the following statement(s) is /are correct regarding the original ammonia sample?

(A) The volume of NH_3 gas taken was 40 ml

(B) The volume of NH_3 gas taken was 50 ml

(C) Only 80% of NH_3 gas decomposed into N_2 and H_2 gases.

(D) Only 20% of NH_3 gas decomposed into N_2 and H_2 gases.

11. When $N_2(g)$ and $H_2(g)$ are mixed N_2H_4 (g) , $NH_3(g)$ or both may form , depending upon the relative amount of N_2 and H_2 taken. If initial moles of N_2, H_2 are x, y and final moles of N_2H_4, NH_3 are z, v, then the correct options from the following in order of (x, y , z, v) is/are -

(A) (2, 2, 1, 0) (B) (3, 8, 1, 4)

(C) (4,9,4,1) (D) (0.5, 3, 0, 1)

SECTION-I (iii) Paragraph : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 12 - 14

Analysis of dissolved oxygen in river water is very essential to ensure that aquatic life is safe. As per european standards dissolved oxygen should never be less than 6.4 mg / lit and preferably should be greater than 9.6 mg / lit. A water body having value less than 6.4 mg / lit. is considered "heavily polluted" and a water body having value greater than 9.6 mg / lit. is considered optimum fit for aquatic life. One method to measure dissolved oxygen is through Winkler titration. The method involves three stages.

Stage-1 : The water sample is treated with excess of a solution of Mn^{2+} (aq) along with a base to cause the following reaction

$Mn^{2+}(aq) + 2OH^{-}(aq) \longrightarrow Mn(OH)_2(s)$ ... (i)

The precipitate then reacts with dissolved oxygen as shown

$2Mn(OH)_2(s) + \frac{1}{2}O_2(aq) + H_2O(\ell) \longrightarrow 2Mn(OH)_3(s)$ ... (ii)

Stage-2 : The solution is added with excess of acidified solution of KI to cause following reaction,

$2Mn(OH)_3(s) + 2KI + 3H_2SO_4 \longrightarrow 2MnSO_4 + K_2SO_4 + I_2(aq) + 6H_2O$ ... (iii)

Stage-3 : The I_2 produced is calculated by the following reaction by calculating amount of $Na_2S_2O_3$ consumed.

$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$ ... (iv)

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Using the above information answer the questions :

12. If 1 litre of water sample is taken then calculate maximum volume of 0.01 M $Na_2S_2O_3$ solution required if the sample is considered "heavily polluted".

(A) 80 ml (B) 80 litre

(C) 120 ml (D) 120 litre

13. If 1 litre of water sample is taken, then calculate minimum volume of 0.1 M $Na_2S_2O_3$ solution required if the sample is considered "optimum fit" for aquatic life.

(A) 8 ml (B) 80 ml

(C) 12 ml (D) 120 ml

14. Which of the statements is correct w.r.t. above method ?

(A) Greater the amount of dissolved oxygen lesser will be moles of $Na_2S_2O_3$ required.

(B) Excess solution in stage-1 ensures all dissolved oxygen reacts.

(C) Moles of $Mn(OH)_2(s)$ reacted and moles of $Mn(OH)_3$ formed will be different.

(D) 1 mole of dissolved oxygen will require 2 moles of $Na_2S_2O_3$ for estimation.

SECTION-II Second Decimal : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

1. 2ml Tetrahedral lead, $(C_2H_5)_4$ Pb, is mixed per litre of gasoline to prevent knocking in engine. How many moles of ethyl chloride, C_2H_5Cl, is needed to produce sufficient tetraethyl lead needed to be mixed with 25 L gasoline ? Density of tetraethyl lead is 6.48 gm / ml. (Pb = 208)
   $4C_2H_5Cl + 4Na - Pb \rightarrow (C_2H_5)_4 Pb + 4NaCl + 3Pb$

2. A sample of $K_2SO_4.Al_2(SO_4)_3.24H_2O$ contains 5 kg oxygen then calculate mass of sulphur (in kg) in same sample?

3. Equivalent weight of Br_2 is 96 in the following disproportionation reaction:
   $Br_2 + OH^- \longrightarrow Br^- + H_2O$ + ? (oxidized product)
   The oxidation state of Br in the oxidized product is (Br = 80)

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. To find molarity of 2 separate solutions of $KMnO_4$ and $O_2(O_2 \rightarrow O^{-2})$, a student titrate 20 ml of $KMnO_4$ (in neutral medium) and 10 ml of O_2 separately using 3 reagents A, B & C , performing 3 different experiments for $KMnO_4$ and O_2 each. He found molarity of $KMnO_4$ and O_2 as $\frac{1}{20}$ M and $\frac{1}{10}$ M respectively.
   A, B & C are $FeCr_2O_4$, $CuFeS_2$, FeS respectively and their reactions are:
   (A) $FeCr_2O_4 \longrightarrow K_2CrO_4 + Fe_2O_3$
   (B) $CuFeS_2 \longrightarrow Cu^+ + Fe^{2+} + SO_2$
   (C) $FeS \longrightarrow Fe^{+3} + SO_2$
   Column-I represents solution of A, B, C used for titration
   Column-II represents mili moles of A, B, C consumed with $KMnO_4$ or with O_2.

	Column I		Column II
(A)	$FeCr_2O_4$	(P)	$\frac{3}{11}$
(B)	$CuFeS_2$	(Q)	$\frac{3}{7}$
(C)	FeS	(R)	$\frac{4}{7}$
		(S)	$\frac{4}{11}$

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SECTION-I (i) Single Correct (Maximum Marks:18)

Each question of this section have (+3, 0) marks.

1. Na - Amalgam is prepared by electrolysis of aq. NaCl solution using 10 gm Hg cathode. How many Faraday's of electricity is required to prepare 18.7% Na-Amalgam. (current Eff. - 50%)

(A) 0.081 F (B) 0.16 F (C) 0.2 F (D) 0.1 F

2. Molar conductances of $BaCl_2$, H_2SO_4 and HCl are x_1, x_2 and x_3 $Scm^2\ mol^{-1}$ at infinite dilution. If specific conductance of saturated $BaSO_4$ solution is of y Scm^{-1}, then k_{sp} of $BaSO_4$ is

(A) $\dfrac{10^3 y}{2(x_1 + x_2 - 2x_3)}$ (B) $\dfrac{10^6 y^2}{(x_1 + x_2 - 2x_3)^2}$

(C) $\dfrac{10^6 y^2}{4(x_1 + x_2 - 2x_3)^2}$ (D) $\dfrac{x_1 + x_2 - 2x_3}{10^6 y^2}$

3. The conductivity of a saturated solution of Ag_3PO_4 is 9×10^{-6} S m^{-1} and its equivalent conductivity is 1.50×10^{-4} S m^2 equivalent^{-1}. The K_{sp} of Ag_3PO_4 is

(A) 4.32×10^{-18} (B) 1.8×10^{-9}

(C) 8.64×10^{-13} (D) None of these

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4. What would be the reduction potential of an electrode at 298 K, which originally contained 1 M $K_2Cr_2O_7$ solution in acidic buffer solution of pH = 1.0 and which was treated with 50% of the Sn necessary to reduce all $Cr_2O_7^{2-}$ to Cr^{3+}. Assume pH of solution remains constant.

Given : $E^0_{Cr_2O_7^{2-}/Cr^{3+},H^+} = 1.33$ V; log 2 = 0.3 ; $\dfrac{2.303\,RT}{F} = 0.06$

(A) 1.285 V (B) 1.193 V

(C) 1.187 V (D) 1.473 V

5. At certain temperature (T) if conductivity of pure water is 5.5×10^{-7} S cm^{-1} then calculate pOH of pure water at temperature T.

Given : $\lambda^{\infty}_{H^+} = 350$ S cm^2 eq^{-1}; $\lambda^{\infty}_{OH^-} = 200$ S cm^2 eq^{-1}

(A) 5 (B) 6 (C) 7 (D) 8

6. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is

$CH_3OH(\ell) + \dfrac{3}{2}O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$

At 298 K standard Gibb's energies of formation for $CH_3OH(\ell)$, $H_2O(\ell)$ and $CO_2(g)$ are -166.2, -237.2 and -394.4 kJ mol^{-1} respectively. If standard enthalpy of combustion of methanol is -726 kJ mol^{-1}, efficiency of the fuel cell will be

(A) 90% (B) 97% (C) 80% (D) 87%

SECTION-I (ii) Multiple Correct : (Maximum Marks: 16)

Each question of this section have (+4, 0) marks.

7. Select the correct statements-

(A) Generally oxidation potential of anodic half cell decreases during reaction

(B) If standard electrode potential of an electrode is greater than zero then it's reduced form is more stable compared to hydrogen gas

(C) $E^0_{Oxd.(H_2O_2/O_2)} = - E^0_{Red.(H_2O_2/H_2O)}$

(D) Inert electrodes provide their surface for oxidation or reduction reactions.

8. The cell potential for the unbalanced chemical reaction

$Hg_2^{2+} + NO_3^- + H^+ \longrightarrow Hg^{2+} + HNO_2 + H_2O$; $E°_{cell} = 0.02$ V

is measured under standard state conditions in the electrochemical cell. $\left(\frac{2.303\ RT}{F} = 0.06\right)$

Then which is/are the correct statement(s) :-

(A) Solution in cathodic compartment is acidic

(B) Solution in Anodic compartment is acidic

(C) Cell potential will be zero at pH = $\frac{2}{9}$, if activity of other components are equal to one.

(D) 0.6 moles of electron pass through the circuit when 0.6 moles of Hg^{2+} are produced in the cell

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9. Pick out the correct statements among the following from inspection of standard reduction potentials

(Assume standard state conditions).

$Cl_2(g) + 2e \rightleftharpoons 2Cl^-(aq.)$ $E°_{Cl_2/Cl^-} = + 1.36$ volt

$Br_2(\ell) + 2e \rightleftharpoons 2Br^-(aq.)$ $E°_{Br_2/Br^-} = + 1.09$ volt

$I_2(s) + 2e \rightleftharpoons 2I^-(aq.)$ $E°_{I_2/I^-} = + 0.54$ volt

$S_2O_8^{2-}(aq.) + 2e \rightleftharpoons 2SO_4^{2-}(aq.)$ $E°_{S_2O_8^{2-}/SO_4^{2-}} = + 2.00$ volt

(A) Cl_2 can oxidise SO_4^{2-} from solution

(B) Cl_2 can oxidise Br^- and I^- from aqueous solution

(C) $S_2O_8^{2-}$ can oxidise Cl^-, Br^- and I^- from aqueous solution

(D) $S_2O_8^{2-}$ is added slowly, Br^- can be reduce in presence of Cl^-

10. Indicate the correct statements -

(A) Conductivity cells have cell constant values independent of solution filled into the cell

(B) DC (direct current) is not used for measuring the resistance of a solution

(C) Kohlrausch law is valid both for strong and weak electrolytes

(D) The k decreases but λ_m and $\lambda_{eq.}$ increase on dilution

SECTION-I (iii) Paragraph : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 11 - 13

In quantitative estimations of any element using an oxidizing agent it is essential to predict which substance (s) gets oxidized by the oxidizing agent. In case more than one substance are getting oxidized then the oxidizing agents gets distributed in all the reactions taking place. From this information and the data given below answer the questions that follow.

$Fe^{+2} \longrightarrow Fe^{+3} + e^-$ $\quad E^o_{O.P.} = -0.77V$

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ $\quad E^o_{R.P.} = 1.51\ V$

$2Cl^- \longrightarrow Cl_2 + 2e^-$ $\quad E^o_{O.P.} = -1.36V$

$2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e^-$ $\quad E^o_{O.P.} = -2.0V$

$C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-$ $\quad E^o_{O.P.} = -0.77V$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ $\quad E^o_{R.P.} = 1.33V$

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11. Which of the following can not be used for quantitative estimation of FeC_2O_4?

(A) $K_2Cr_2O_7$ in presence of dil H_2SO_4.

(B) $K_2Cr_2O_7$ in presence of dil. HCl

(C) $KMnO_4$ in presence of dil. HCl.

(D) $KMnO_4$ in presence of dil. H_2SO_4

12. Millimoles of FeC_2O_4 in the solution if 50 ml of 0.1 M $K_2Cr_2O_7$ is used for its oxidation in presence of dil. H_2SO_4.

(A) 10 millimoles

(B) 30 millimoles

(C) 20 millimoles

(D) quantitative estimation can't be made from given data

13. Millimoles of FeC_2O_4 in the solution, if 50 ml of 0.1 M $KMnO_4$ is used for its oxidation in presence of dill. HCl if 3.5 millimoles of Cl_2 is obtained along with other products.

(A) 25/3 millimoles

(B) 6 millimoles

(C) 12.5 millimoles

(D) quantitative estimation can't be made from given data

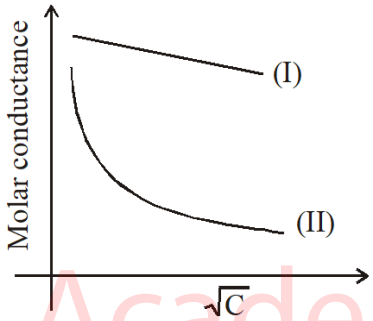
SECTION-II Second Decimal : (Maximum Marks: 15)

Each question of this section have (+3, 0) marks.

1. Calculate the magnitude of electrode potential in volt having reaction: $Ag_2S\ (s) + 2e^- \rightleftharpoons 2Ag(s) + S^{2-}(aq)$. in a solution buffered at pH = 3 and which is also saturated with 0.1 M H_2S (aq).
Given For H_2S : $K_{a_1} = 10^{-7}$ and $K_{a_2} = 10^{-14}$, $K_{sp}(Ag_2S) = 10^{-49}$, $E^0_{Ag^+/Ag} = 0.8$.
Use $2.303 \times \frac{RT}{F} = 0.06$

2. A current of 0.1 amp is passed though excess of molten mixture of Al_2O_3 and Na_3AlF_6 for 96.5 hours. The mass of Al (in gm) deposited at the cathode, with $\frac{1000}{10.8}$% current efficiency is (Al = 27).

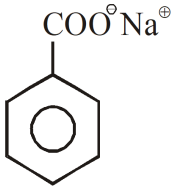
3. Given graph was plotted between the molar conductance of various electrolytes and $\sqrt{C}$ generally (I) & (II) types of graphs are obtained.



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Among the following the number of species having similar graph to the (II) is

CH_3COOH,  , KCl, HCl,  ,

$AgNO_3$, CH_3COONa, Na_2HPO_3, H_3PO_4, 

4. For an element 'X' the process of oxidation is :
$X_2O_4^{-2} \longrightarrow$ New compound
If 965 A current when passed for 100 seconds discharged 0.1 mol of $X_2O_4^{-2}$, find oxidation state of X in new compound ?

5. Calculate e.m.f of cell (in volts)
$Ag \mid AgNO_3\ (0.1M) \parallel Ag(NH_3)_2^+\ (0.01\ M), NH_3(1M) \mid Ag$
Given : - $K_f([Ag(NH_3)_2^+]) = 10^8$
$\frac{2.303RT}{F} = 0.06$

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. Match the following Column–I and Column–II :

	Column–I		Column–II
(A)	Metal–metal ion half cell	(P)	Cl^- | AgCl | Ag
(B)	Gas–gas ion half cell	(Q)	Pt | Cr^{2+} (C_1M) , Cr^{3+} (C_2M)
(C)	Redox half cell	(R)	Pt | O_2 | OH^-
(D)	Metal insoluble metal salt-anion half cell	(S)	Sn^{2+} | Sn
		(T)	Pt | Sn^{4+} , Sn^{2+}

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SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have (+3, 0) marks.

1. For the reaction : $xA \rightarrow yB$,

$$\log_{10}\left(-\frac{d[A]}{dt}\right) = \log_{10}\left(+\frac{d[B]}{dt}\right) + 0.3$$

If the value of $\log_{10} 5 = 0.7$, the value of x : y is :

(A) 2 : 1 (B) 1 : 2

(C) 3 : 10 (D) 5 : 2

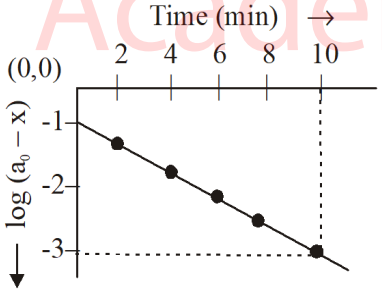
2. For acid catalysed hydrolysis of ester rate law obtained is rate = k [ester] $[H^+]$, where k = 0.01 M^{-1} hr^{-1}. What is the half-life if the initial concentrations are 0.02 M for the ester and 0.05 M for the catalyzing acid.

(A) 1429 hours (B) 5000 hours

(C) 69.3 hours (D) 1386 hours

3. For the first order decomposition of $SO_2Cl_2(g)$, $SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$ a graph of $\log (a_0 - x)$ vs t is shown in figure. What is the rate constant (sec^{-1})?

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(A) 0.2 (B) 4.6×10^{-1}

(C) 7.7×10^{-3} (D) 1.15×10^{-2}

4. If 0.01 % of a substance undergoing decomposition is consumed in millisecond when the concentration is 0.02 M and in 0.25 millisecond when the concentration is 0.04 M, what is the order of the reaction?

(A) 3 (B) 2 (C) 1 (D) zero

5. A first order reaction : $A_{(g)} \rightarrow nB_{(g)}$, is started with pure A. The reaction takes place at constant volume and temperature . If the initial pressure of system was P_0 and the rate constant of reaction is 'K' min^{-1}, then the total pressure of the system at t = t min, is :

(A) $P_0 . e^{-Kt}$ (B) $n . P_0 \left(1 - e^{-K1}\right)$

(C) $P_0 \left[1 + n . \left(1 - e^{-Kt}\right)\right]$ (D) $P_0[n+(1- n).e^{-Kt}]$

6. C^{14} in a fossil of 'X' is presently 1/5 times of that in present atmosphere. Given that due to nuclear explosion after death of 'X' there was 25% increase in C^{14} content in atmosphere. The correct age of fossil is $x \times 10^4$ yrs, then find the value of x.
Given : $T_{1/2}$ for C^{14} = 6000 yrs.

(A) 1.20 (B) 1.6 (C) 2.4 (D) 1.8

7. Calculate the value of Activation energy at temperature of 300 K if the expression of rate constant as a function of temperature is given as $K = 30e^{\frac{-40}{T}} . e^{\frac{300}{T^2}}$.

(A) 40 R (B) 80 R

(C) 300 R (D) 38 R

8. The radioactive decay $_{83}Bi^{211} \longrightarrow {}_{81}Tl^{207}$, takes place in 100 L closed veseel at 27°C. Starting with 2 moles of $_{83}Bi^{211}$ ($t_{1/2}$ = 130 sec), the pressure development in the vessel after 520 sec will be -

(A) 1.875 atm (B) 0.2155

(C) 0.4618 atm (D) 4.618 atm

SECTION-I (ii) Multiple Correct : (Maximum Marks: 20)

Each question of this section have (+4, 0) marks.

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9. Which of the following is/are correct statement(s)?

(A) Stoichiometry of a reaction tells about the order of the elementary reactions.

(B) For a zero order reaction, rate and the rate constant are identical.

(C) A zero order reaction is controlled by factors other than concentration of reactants.

(D) A zero order reaction is always elementary reaction.

10. The activation energies of two reactions I and II are E_a and $2E_a$ respectively. If the temperature of the reacting systems is increased from T to T', predict which of the following alternative is/are correct ?
(k represent rate constant)

(A) $k'_I / k_I > 1$ (B) $k'_{II} / k_{II} > 1$

(C) $k'_I / k_I < k'_{II} / k_{II}$ (D) $k'_I / k_I = 2k'_{II} / k_{II}$

11. Select the correct option (s)

(A) Specific activity of U^{235} is greater than specific activity of U_3O_8 (All 'U' is U^{235})

(B) Activity of 1 mol U^{235} is greater than activity of 1mol U_3O_8 (All 'U' is U^{235})

(C) Decay constant of U^{235} is same whether it is pure U^{235} or U_3O_8 (All 'U' is U^{235})

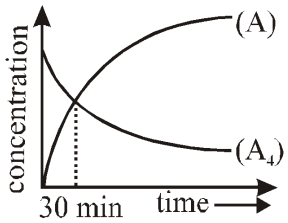
(D) Rate of radioactive disintegration increases with increase in temperature

12. Consider the following first order decomposition reaction,

$A_4(g) \longrightarrow 4A(g)$

Which of the following statements regarding the reaction are correct?

[Given : log 2 = 0.30]



(A) At 30 min, only 20% reaction is complete

(B) $t_{1/2}$ of reaction is approximately 90 min.

(C) Rate of reaction decreases linearly with time .

(D) The time for intersection of two curves is independent of initial concentration of A_4.

13. Rate of radioactive decay for a radioactive element depends on :

(A) Amount of radioactive element

(B) Temperature

(C) Pressure

(D) Isotopic form of radioactive element

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SECTION-I (iii) Paragraph : (Maximum Marks: 15)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 14 and 15

For a zero order chemical reaction,

$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g)$

rate of reaction = 0.1 atm/sec. Initially only NH_3 is present & Its pressure = 3atm.

14. Rate of disappearance of NH_3 will be

(A) 0.1 atm/sec.

(B) 0.2 atm/sec.

(C) 0.05 atm/sec.

(D) 0.04 atm/sec

15. Total pressure at t = 10 sec, will be

(A) 4 atm

(B) 5 atm

(C) 4.5 atm

(D) 3.5 atm

Paragraph for Questions 16 - 18

A substance 'X' dissociates by two parallel reactions with different ' Arrhenius constants (A_1 and A_2)" and different "Activation energies" (E_{a_1} and E_{a_2}) as shown.

$$X(g) \xrightarrow[E_{a_1} = 12K\ cal/mol]{A_1 = 2 \times 10^{14} s^{-1}} Y(g) \quad ; \quad X(g) \xrightarrow[E_{a_2} = 9K\ cal/mol]{A_2 = 2 \times 10^{14} \times e^{-2} s^{-1}} Z(g)$$

[Given : R = 2 Cal/mol-K]

16. Temperature at which net activation energy for dissociation of X will be 10.5 KCal/mol?

(A) 200 (B) 750 (C) 800 (D) 1000

17. Which of the following statements are correct with respect to major product obtained in the above parallel decay:

(A) "Y" will be the major product at temperature 300 K.

(B) "Z" will be the major product at temperature 800 K.

(C) Both statements are correct

(D) None of the statements are correct

18. If the temperature is increased, then

(A) the rate of formation of Y will increase but the rate of formation of Z will decrease

(B) the rate of formation of both Y and Z will increase, but the percentage increase in rate will be higher for the formation of Y.

(C) the rate of formation of both Y and Z will increase but the percentage increase in rate will be higher for the formation of Z.

(D) the rate of formation of both Y and Z will increase, and the percentage increase in the rate will be same for both.

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SECTION-II Second Decimal : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

1. A radioactive isotope disintegrates and emits 6 β and 7 α particles to form a stable product. Find the difference between atomic number of parent radioactive isotope and the stable product.

2. An optically active substance 'A' is decomposing into optically active substance 'B' and 'C' as

$$A \xrightarrow{K=10^{-3} min^{-1}} 2B + C$$

The specific rotations of A, B and C are + 40° , + 10° and −30° per mole, respectively. Initially 'A' and 'C' were present in 4 : 3 mole ratio. Close to how many hours the sample becomes optically inactive ?

[Given : $\ell n\frac{7}{5} = 0.34$, $\ell n\frac{20}{13} = 0.42$]

3. Kinetic data for hydrolysis of a 0.3 M solution of ethyl acetate in presence of HCl as catalyst is given in the following table. A constant volume of reaction mixture is taken at different time and titrated against standard alkali solution at different time.

Time/min	0	10	$t = \infty$
Vol. of Alkali/ml	25.00	28.00	40.00

It the rate law is given by rate = $k[Ester]^1[H^+]^1$, where H^+ is a catalyst, find the rate (in M/min) of ester hydrolysis in a solution which is 0.1 M each in ester and acid HCl (M/min)

Use: (ln 2 = 0.7, ln5 = 1.6)

If your answer is $x \times 10^{-y}$ then fill 'x' in OMR sheet.

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. Match the Column :

	Column-I		**Column-II**
(A)	Molecularity of an elementary step reaction	(P)	Can never be –ive
(B)	Order of an elementary step	(Q)	Can be fractional
(C)	Order of a reactant in composite rate law	(R)	Is always natural number
(D)	Rate constant	(S)	Fixed value at constant temperature and reaction conditions

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SECTION-I (i) Single Correct (Maximum Marks:27)

Each question of this section have (+3, 0) marks.

1. A mixture of A and B in the molar ratio 1 : 2 forms a maximum boiling azeotrope. Identify the incorrect statement, if A is more volatile [Molar mass of A = 100, Molar mass of B = 50]

(A) A liquid solution of A & B having mass % of A = 50 will have vapours having mass % of A = 50.

(B) A liquid solution A and B having mass % of A > 50% will have vapours having mass % of A > 50%.

(C) A mixture of A & B in the molar ratio 1 : 3 can be separated into azetropic mixture and pure A.

(D) A mixture of A and B in the molar ratio 2 : 3 can be separated into azeotropic mixture and pure A.

2. Air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298K, if the Henry's law constants for oxygen and nitrogen are 3.30×10^7 mm and 6.51×10^7 mm, respectively, the mole ratio of gases (O_2 : N_2) dissolved in water is about

(A) 1 : 2 (B) 2 : 1 (C) 1 : 8 (D) 1 : 3

3. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution at 100°C is -

(Take $K_b = 0.76$ K kg mol^{-1})

(A) 724 (B) 740 (C) 736 (D) 718

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4. 20 moles of liquid A are mixed with 20 moles of liquid B to form an ideal binary solution. Calculate moles of A in liquid state when half of the solution has vaporized.

(Given : $P_A^0 = 100$ torr, $P_B^0 = 121$ torr)

(A) $\frac{11}{21}$ (B) $\frac{110}{21}$ (C) $\frac{200}{21}$ (D) $\frac{220}{21}$

5. The freezing point depression of a 0.10 M solution of formic acid is -0.2046°C. What is the equilibrium constant for the reaction at 298 K?

$HCOO^-(aq) + H_2O(\ell) \rightleftharpoons HCOOH(aq) + OH^-(aq)$

(Given : $K_f(H_2O) = 1.86$K kg mol^{-1} , Molarity = molality)

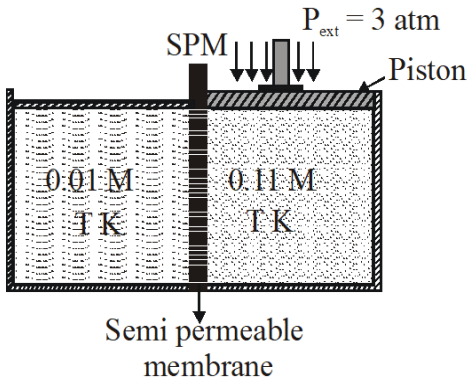
(A) 1.11×10^{-3} (B) 9×10^{-12} (C) 9×10^{-13} (D) 1.1×10^{-11}

6. 1kg of water is mixed with Sucrose and is cooled and maintained at –4°C. How much ice will be separated out if the moles of the sucrose is 0.75 in the solution ? $K_f(H_2O) = 1.86$ Kg mol^{-1}K.

(A) 578.5 gm (B) 534.25 gm (C) 651.25 gm (D) 561.25 gm

7. A total pressure of 3 atm when applied on the surface of solution of concentration 0.11 M prevent flow of solvent across semipermeable membrane from other side of solution of concentration of 0.01 M, as shown in the diagram. What is the experimental temperature ?

(Take : R = 0.08 L-atm/K-mole)



(A) 375 K (B) 250 K

(C) 125 K (D) 225 K

8. 1 litre of urea solution (1M) is mixed with 2 litre of 0.3M NaCl solution both at TK, osmotic pressure would be -

(A) 0.73 RT (B) 0.8 RT

(C) 0.65 RT (D) 0.5 RT

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9. Which of the following gas has lowest K_H for dissolution in H_2O at 25°C

(A) SO_2 (B) Ne

(C) Ar (D) Kr

SECTION-I (ii) Multiple Correct : (Maximum Marks: 16)

Each question of this section have (+4, 0) marks.

10. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult law. The normal boiling point (T_b) of pure acetone is less than that of pure CS_2. Pick out the incorrect statement(s) among the following.

(A) Boiling temperature of mixture is always less than boiling temperature of acetone.

(B) Boiling temperature of Azeotropic mixture is always less than boiling temperature of pure CS_2.

(C) When a small amount of CS_2 (less volatile component) is added to excess of acetone boiling point of resulting mixture increases.

(D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation.

11. Blood cell s in the human body have semipermeable membrane & depending upon concentration of solution inside blood cells & outside (in the blood); " Lysis" [expansion of blood cells] & " Crenatin"[Contraction of blood cells] may occur. Kidneys are responsible for keeping solution inside blood cell & blood at same concentration. Identify the options which is /are correct?

(A) Lysis will occur when blood cells are kept in a solution which is isotonic with blood

(B) Crenatin will occur when blood cells are kept in a solution which is hypertonic with blood.

(C) Blood cells will have normal shape when placed in a solution isotonic with blood.

(D) Lysis occurs when blood cells are kept in hypotonic solution as compared to blood.

12. For an ideal binary liquid solution with $P_A^0 > P_B^0$, which relation between X_A (mole fraction of A in liquid phase) and Y_A(mole fraction of A in vapour phase) is correct?

(A) $Y_A < Y_B$

(B) $X_A > X_B$

(C) $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$

(D) $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$

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13. A non - volatile solute 'X' completely dimerises in water, if the temperature is below $-3.72°C$ and the solute completely dissociates as $X \longrightarrow Y + Z$, if the temperature is above 100.26°C. In between these temperture, (including both temperature) 'X' is neither dissociated nor associated. One mole of 'X' is dissolved in 1 kg water.

Given: K_b of water $= 0.52$ K. Kg mol^{-1}

K_f of water $= 1.86$ K. Kg mol^{-1}

Which of the following statement(s) is/are true regarding the solution?

(A) The freezing point of solution is $-1.86°C$

(B) the boiling point of solution is 101.04°C

(C) When the solution is cooled to $-7.44°C$, 75% of water present initially will separate as ice.

(D) When the solution is heated to 102.08°C, 50% of water present initially will escape out as vapour.

SECTION-I (iii) Paragraph : (Maximum Marks: 18)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 14 - 16

The lowering of vapour pressure on adding a non-volatile solute may be measured by Ostwald- Walker method, In this method, dry air is first passed through a series of vessels having the solution, then through a series of vessels having pure solvent and finally through a vessel (normally U-tube) having the absorbent of the solvent. The masses of solution and pure solvent decreases due to removal of vapours of solvent in the flow of air and the mass of absorbent increases due to absorption of the vapour of solvent, in order to make the air dry. By measuring the changes in mass, we may determine the lowering of vapour pressure and hence, the composition of solution, form Raoult['s law.

14. If the mass of absorbent increased by 0.24 gm and the mass of pure solvent (water) decreased by 0.02 gm , then the mass percent of solute (glucose) in its aqueoue solution is :

(A) $\frac{1000}{21}\%$

(B) $\frac{100}{12}\%$

(C) $\frac{1000}{22}\%$

(D) $\frac{100}{11}\%$

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15. If the experiment is performed with aq. $AlCl_3$ solution ($\alpha = 0.8$) prepared by dissolving 1 mole $AlCl_3$ in 17 mole water and the decrease in mass of solution is found to be 0.18 gm, then the increase in mass of absorbent should be :

(A) 0.216 gm

(B) 0.225 gm

(C) 0.191 gm

(D) 1.08 gm

16. If the arrangement of vessels is changed in the experiment as :

Dry air $\xrightarrow{\text{Pure solvent}}$ $\xrightarrow{\text{solution}}$ $\xrightarrow{\text{Absorbent}}$ Dry air.

then, which of the following must be incorrect?

(A) the mass of pure solvent will decrease.

(B) the mass of solution will decrease.

(C) the mass of absorbent will increase.

(D) The experiment will obey the law of conservation of mass.

Paragraph for Questions 17 - 19

One mole of liquid 'A' and two moles of liquid 'B' are mixed 27°C to form an ideal binary solution. (ln2 = 0.7, ln3 = 1.1)

17. The value of ΔH_{mix} is

(A) 0 (B) + ve

(C) – ve (D) any of these

18. The value of ΔG_{mix} is

(A) 0 (B) + 1140 cal / K

(C) – 1140 cal / K (D) – 380 cal / K - mol

19. The value of ΔS_{mix} is

(A) 0 (B) + 3.8 cal / K

(C) – 3.8 cal / K (D) + 3.8 cal / K - mol

SECTION-II Second Decimal : (Maximum Marks: 12)

Each question of this section have (+3, 0) marks.

1. If 1 gm of solute dimerise upto 75% in 100 gm of H_2O & show depression in freezing point 0.093°C. Find average molecular weight of solute in the solution $\left(K_{f(H_2O)} = 1.86\right)$.

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2. 4ml of a gas at 1 atm pressure and 300 K is dissolved in 1L of solution. Calculate the volume of the gas dissolved at 4 atm pressure and 300 K in $\frac{1}{2}$L of solution in ml.

3. In how many of the following, the elevation in boiling point of the final solution is more than that of initially taken solution (No limiting reagent) (Assume $\alpha = 1$ where ever needed)

(i) $BaCl_2(aq.) + Na_2SO_4(aq.)$ (added) $\longrightarrow BaSO_4 + 2NaCl(aq.)$

(ii) $K_4Fe(CN)_6(aq.) + 2CuSO_4(aq.)$ (added) $\longrightarrow Cu_2Fe(CN)_6 + 2K_2SO_4$

(iii) $Al_2(SO_4)_3(aq.) + 3BaCl_2$ (aq.) (added) $\longrightarrow 3BaSO_4 + 2AlCl_3(aq.)$

(iv) $AgNO_3$ (aq.) + 2KCN (added) $\longrightarrow K[Ag(CN_2)(aq.) + KNO_3(aq.)$

4. When the mixture of two immicible liquids (water and nitrobenzene) boils at 372 K and the vapour pressure at this temperature are 98.4 kPa (H_2O) and 3.6 kPa ($C_6H_5NO_2$). If the weight ratio of nitrobenzene and water in the vapour is 1 : x, then the value of 'x' is.

SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have (+3, 0) marks.

1. If 250 mL of N_2 over water at 30°C and a total pressure of 740 torr is mixed with 300 mL of Ne over water at 30°C and a total pressure of 780 torr, what will be the total pressure if the mixture is in a 500 mL vessel over water at 30°C

(Given : Vapour pressure (Aqueous tension) of H_2O at 30°C is 40 torr. Assume volume of $H_2O(\ell)$ is negligible in final vessel)

(A) 760 torr (B) 820 torr

(C) 780 torr (D) 824 torr

2. If the number of molecules of SO_2 (M = 64) effusing through an orifice of unit area of cross section in unit time at 0°C and 1 atm pressure is 'n'. The number of He molecules (M = 4) effusing in unit time under similar conditions at 273°C and 0.25 atm is :

(A) $\frac{n}{\sqrt{2}}$ (B) $\frac{n}{2}$

(C) 2n (D) $\sqrt{2}\,n$

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3. The most probable kinetic energy of molecules in a sample of ideal gas is

(A) KT (B) $\frac{1}{2}KT$

(C) $\frac{3}{2}KT$ (D) $\frac{1}{3}KT$

4. An open vessel at 27°C is heated untill 3/8th of air in it has been expelled. Assuming that the volume remain constant, find the temperature at which the vessel was heated :

(A) 307°C (B) 107°C

(C) 480°C (D) 207°C

5. The density of a gaseous substance at 1 atm pressure and 773 K is 0.4 g/L. If the molecular weight of the substance is 30, then dominant forces existing among gas molecules is :

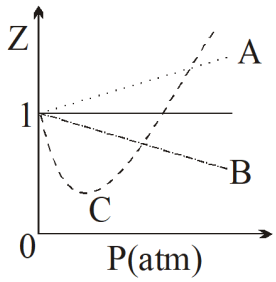
(A) repulsive (B) attractive

(C) both (D) none

6. 0.5 mol of a diatomic gas present in 10 L vessel at certain temperature exert a pressure of 0.96 atm. Under similar conditions an ideal gas exerted 1.0 atm pressure. If volume of gas molecules is negligible then, the value of vander waals constant "a" (in atm L^2/mol^2) is :

(A) 4 (B) 16 (C) 32 (D) 8

7. The given graph represents the variation of Z(compressibility factor = $\frac{PV}{nRT}$) versus P, for three real gases A, B and C. Identify the only incorrect statement.



(A) for the gas A, a = 0 and its dependence on P is linear at all pressure

(B) for the gas B, b = 0 and its dependence on P is linear at all pressure

(C) for the gas C, which is typical real gas for which neither a nor b = 0. By knowing the minima and the point of intersection, with Z = 1, a and b can be calculated.

(D) At high pressure, the slope is positive for all real gases A, B and C.

8. One litre of a gas at 300 atm and 473 K is compressed to a pressure of 600 atm and 273 K. The compressibility factors are 1.072 and 1.375 respectively at initial and final states. What will be the final volume :

(A) 307.8 mL

(B) 370.1 mL

(C) 22.4 mL

(D) 740.2 mL

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SECTION-I (ii) Multiple Correct : (Maximum Marks: 16)

Each question of this section have (+4, 0) marks.

9. Select the correct statement for a real gas

(i) The behaviour of the gas become more ideal when pressure is very low.

(ii) Pressure of gas for ideal behaviour depends upon nature of the gas and its temperature.

(iii) The temperature at which a real gas obeys ideal gas law over an appreciable range of pressure in low pressure range is called Boyle's temperature.

(A) Only (i), (ii)

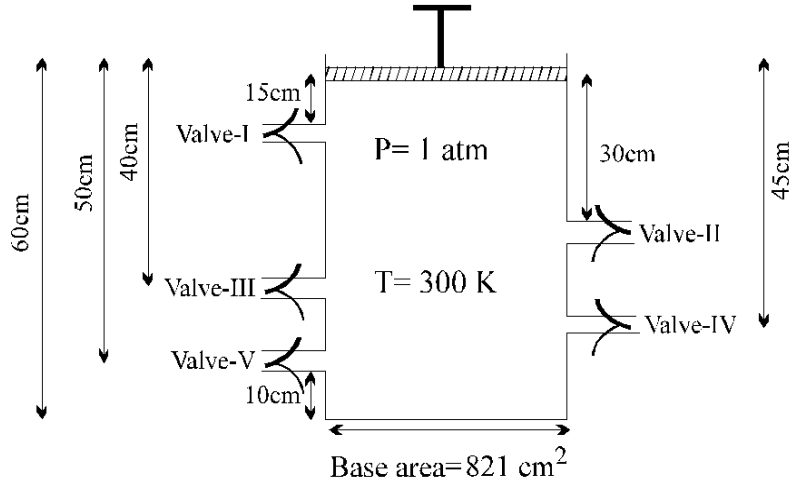
(B) Only (i)

(C) Only (ii), (iii)

(D) (i), (ii), (iii)

10. A container fitted with frictionless massless piston consist of five valves–I, II, III, IV and V. These valves open automatically if pressure exceed over 1.5, 2.2, 2.5, 4.4 and 4.8 atm respectively. Under the given initial conditions (mentioned in given diagram) system is in state of equilibrium. Piston is now pressed in downward direction very slowly.

[Note: Consider the diameter of valve tube negligible and temperature remain constant.]



Select the correct option(s):

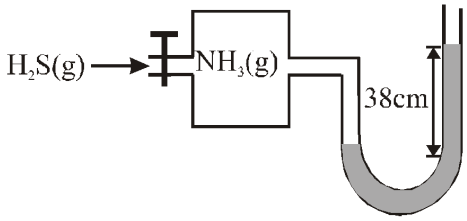
(A) Valve –II will be opened first

(B) As the piston crosses the valve which will be opened first, the remaining number of moles in container are $\frac{5}{3}$.

(C) Valve–V will be the second valve which open

(D) Number of moles will zero as piston crosses Valve–V

11. A closed container of volume 4.92 litre contains NH_3 gas at 27°C as shown:



H_2S gas is passed inside the container at a rate of 0.01 mole/sec, such that no gas is coming out from the container. The reaction take place in the container is

$$2NH_3(g) + H_2S(g) \xrightarrow{100\%} (NH_4)_2S(s)$$

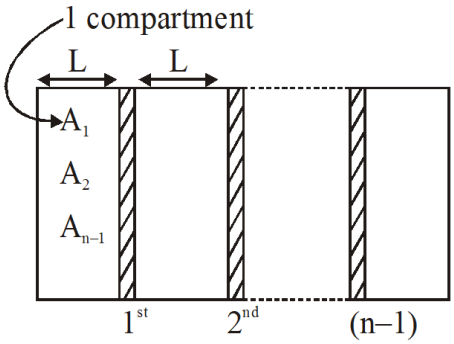
(Assume that there is no time lag in between H_2S gas passed and $(NH_4)_2$ S formed.)

The time at which the difference in Hg column is equal to 38 cm.

(A) 0 sec. (B) 20 sec. (C) 10 sec (D) 25 sec.

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12. A very long rectangular box is divided into 'n' equal compartments with (n – 1) fixed SPM (semi permeable membrance) numbered from 1 to (n – 1) [n is unknown] as shown. The gases are initially present in only 1 compartment & all gases have same moles & after substantial time ratio of P_{A_4} in 3rd compartment to $P_{A_{n-1}}$ in 1st compartment is 3 (where P_{A_4} & $P_{A_{n-1}}$ represents partial pressure of A_4 & A_{n-1} respectively).



Select the correct option(s) based on given condition.
Condition-1 : Gases can pass through only those SPM whose number is less than to their subscript (like A_2 can pass through 1st SPM only, A_3 through 1st & 2nd & soon).
Condition-2 : Gases can pass through only those SPM whose number is less than or equal to their subscript (like A_1 can pass through 1st SPM only, A_2 through 1st & 2nd & so on).

(A) The value of n is 15 if condition-2 is followed

(B) The value of n is 13 if condition-1 is followed

(C) After substantial time ratio of n_{A_3} in 2nd compartment to n_{A_5} in 4th compartment is 1.5 if condition-2 is followed (where n_{A_3} and n_{A_5} are number moles of A_3 and A_5 moles respectively)

(D) After substantial time ratio of P_{A_6} and in 6th compartment to P_{A_1} in 1st compartment is 2/7 if condition-1 is followed

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SECTION-I (iii) Paragraph : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 13 - 15

Recent communications with the inhabitants of Neptune have revealed that they have a Celsius type temperature scale, but based on the melting point (0°N) and boiling point (100°N) of their most common substance, hydrogen. Further communications have revealed that the Neptunians know about perfect gas behaviour and that they find in the limit of zero pressure, the value of PV_m is 28L - atm at 0°N and 40L - atm at 100°N.

13. What is the value of the absolute zero of temperature on their temperature scale?

(A) 233.33°C (B) – 233.33°C (C) – 233.33°N (D) 233.33°N

14. What is the value of universal gas constant?

(A) 0.082L - atm/ K - mol (B) 0.12L- atm/ K- mol

(C) 0.4L - atm/K-mol (D) infinite

15. What is the volume occupied by 2 mole of a perfact gas at 2 atm and 66.67°N?

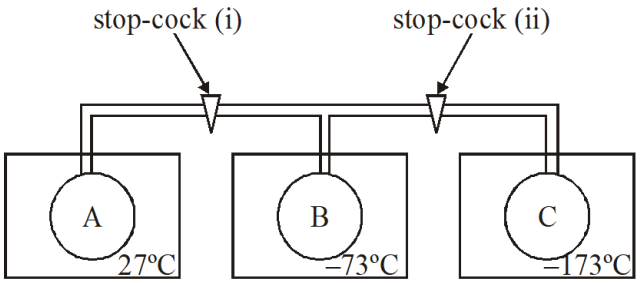
(A) 36 L (B) 24.6 L (C) 20.0 L (D) 8.0 L

SECTION-II Second Decimal : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

1. 500 ml of a vessel contains an ideal gas saturated with water vapours at 3.2 atm and 47ºC. Now the entire mass is transferred into vessel of 1800 ml capacity maintained at 87°C. The final pressure (in atm) becomes (vapour pressure of water at 47º and 87º C are 0.2 atm and 0.6 atm, respectively)

2.



The above apparatus consists of three temperature jacketed 12.315 litre bulbs connected by stop cocks. Bulbs 'A' consist of $NH_{3(g)}$, $CO_{2(g)}$ & $N_{2(g)}$ at 27°C and a total pressure of 60 atm. Bulb 'B' and 'C' are empty and maintained at temperature of –73°C and –173°C respectively. Initially stop-cocks are closed and volume of the connecting lines is zero.

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Given :

(i) $NH_{3(g)}$ liquifies at –20°C, $CO_{2(g)}$ liquifies at –80°C and $N_{2(g)}$ liquifies at –180°C.

(ii) Vapour pressure of $NH_{3(\ell)}$ at –73°C = 1 atm

(iii) Vapour pressure of $NH_{3(\ell)}$ at –173°C is negligible

(iv) Vapour pressure of $CO_{2(\ell)}$ is negligible

(v) On opening stop-cock (i) only, the pressure in containers becomes 17 atm and, on opening both stop-cocks pressure decreases to 4 atm

what is the value of $(n_{NH_3} + n_{CO_2} - n_{N_2})$ in the initial mixture ?

3. A barometer reads 75 cm Hg, when the atmospheric pressure is 80 cm Hg. The same barometer reads 84 cm Hg, when the atmospheric pressure is 90 cm Hg. What should be the reading of barometer when the atmospheric pressure is 70 cm Hg. (Given : $\sqrt{199^2 - 4 \times 8760} = 68$)

If the answer is x cm Hg, fill OMR the value of $\left(\frac{8x}{131}\right)$.

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

	Column I		Column II
(A)	Hydrogen gas (P = 200 atm, T = 273 K)	(P)	Compressibility factor ≠ 1
(B)	Hydrogen gas (P ~ 0, T = 273 K)	(Q)	Attractive forces are dominant
(C)	CO_2 (P = 1 atm, T = 273 K)	(R)	PV = nRT
(D)	Real gas with very large molar volume	(S)	P (V– nb) = nRT

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SECTION-I (i) Single Correct (Maximum Marks:24)

Each question of this section have (+3, 0) marks.

1. It was found that in 30 unit cells of ZnS, four atoms are of trivalent metal. How many vacant tetrahedral voids are present in 30 unit cells of ZnS?

(A) 2 (B) 62 (C) 122 (D) 126

2. A solid element (monoatomic) exist as cubic crystal. If its atomic radius is 1.0 Å and the ratio of packing fraction and density is 0.1 cm^3/gm, then the atomic mass of the element is ($N_A = 6 \times 10^{23}$)

(A) 8π (B) 16π (C) 80π (D) 4π

3. A metal crystallises in such a lattice that only 70% of the total space crystal is occupied by the atoms. If the atomic mass of metal is 32π gm mol^{-1} and atomic radius is 0.2 nm, the density of the metal is :

(A) 7.0 gm/cm^3 (B) 3.5 gm/cm^3 (C) 10.5 gm/cm^3 (D) 14.0 gm/cm^3

4. Cesium iodide consists of a simple cubic lattice of I^- ions with Cs^+ ions in the cubic holes. If the cell edge length is 0.445 nm, what is the Cs–I interatomic distance ?

(A) 0.223 nm (B) 0.314 nm (C) 0.385 nm (D) 0.770 nm

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5. The ZnS (zinc blende) structure is cubic. The unit cell may be described as a face-centered sulfide ion sublattice with zinc ions in the centers of alternating minicubes made by partitioning the main cube into 8 equal parts. How many nearest neighbors does each Zn^{2+} have and what angle is made by the lines connecting any Zn^{2+} to any two of its nearest neighbors?

(A) 4 , 90º (B) 6 , 90º (C) 4 , 109.5º (D) 6 , 109.5º

6. Which of the following oxides behaves as conductor or insulator depending upon temperature?

(A) TiO (B) SiO_2 (C) TiO_3 (D) MgO

7. Which of the following oxides shows electrical properties like metals?

(A) SiO_2 (B) MgO

(C) $SO_2(s)$ (D) CrO_2

8. A compound has cubical unit cell in which X atoms are found at 6 corners and Y atoms are present at remaining corners and at face center which are not opposite to each other and Z-atoms are present at body center and remaining face center. Calculate density in amu/$(A°)^3$, if atomic mass of X, Y and Z are 20, 40 and 60 amu respectively and side of unit cell (2Å).

(A) $\frac{235}{8}$ (B) $\frac{940}{8}$

(C) $\frac{470}{8}$ (D) $\frac{235}{16}$

SECTION-I (ii) Multiple Correct : (Maximum Marks: 16)

Each question of this section have (+4, 0) marks.

9. The incorrect statement(s) for the packing of identical spheres in two dimension is :

(A) For square close packing , coordination number is 4.

(B) For hexagonal close packing, coordination number is 6.

(C) There is only one void per atom in both, square and hexagonal close packing.

(D) Hexagonal close packing is more efficiently packed than square close packing.

10. Which of the following statements are correct?

(A) Ferrimagnetic substances lose ferrimagnetism on heating and become paramagnetic.

(B) Ferrimagnetic substances do not lose ferrimagnetism on heating and remain ferrimagnetic.

(C) Antiferromagnetic substances have domain structures similar to ferromagnetic substances and their magnetic moments are not cancelled by each other.

(D) In ferromagnetic substances all the domains get oriented in the direction of magnetic field and remain as such even after removing magnetic field.

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11. If observed density of crystal is less than ideal density, then the defect may be -

(A) Schottky defect

(B) Frenkel defect

(C) substitutional impurity

(D) metal deficiency defect

12. The co-ordination number of FCC structure for metals is 12, since

(A) each atom touches 4 others in same plane, 3 in plane above and 3 in plane below.

(B) each atom touches 4 others in same plane, 4 in plane above and 4 in plane below.

(C) each atom touches 6 others in same layer, 3 in layer above and 3 in layer below.

(D) each atom touches 3 others in same layer, 6 in layer above and 6 in layer below.

SECTION-I (iii) Paragraph : (Maximum Marks: 15)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 13 and 14

Spinel is an important class of oxide having two types of metal ions. A type of spinal have ccp arrangement to O^{2-} ions in which Fe^{2+} cation occupy $1/8^{th}$ of tetrahedral voids and Fe^{3+} cation occupies half of the octahedral voids. If 'a' be the edge of unit cell then.

13. What is formula of spinel -

(A) Fe_2O_3 (B) Fe_3O_4

(C) FeO (D) Fe_4O_5

14. What is the minimum internuclear distance between Fe^{2+} ion & Fe^{3+} ion in spinel -

(A) $\frac{\sqrt{3}a}{4}$ (B) $\frac{a}{2}$

(C) $\frac{a}{\sqrt{2}}$ (D) a

Paragraph for Questions 15 - 17

	Column-I		Column-II		Column-III
(A)	Hexagonal	(P)	$a = b \neq c$	(I)	Primitive
(B)	Orthorhombic	(Q)	$a \neq b \neq c$	(II)	Body centred
(C)	Monoclinic	(R)	$\alpha = \beta \neq \gamma$	(III)	Face centered
(D)	Triclinic	(S)	$\alpha \neq \beta \neq \gamma$	(IV)	End centered

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15. Which of the following is only correct match :

(A) A, Q, II (B) C, R, II

(C) D, S, II (D) A, P, I

16. Which of the following is only incorrect match :

(A) A, P, I (B) B, P, IV

(C) C, R, I (D) D, S, I

17. Which of the following is only correct match :

(A) A, P, II

(B) C, Q, III

(C) D, S, IV

(D) B, Q, I

SECTION-I (iv) Matching List : (Maximum Marks: 3)

Each question of this section have (+3, 0) marks.

18.

List-I Type of ionic compound		List-II	
(P)	NaCl type	(A)	Cation at 50% of tetrahedral void
(Q)	ZnS type	(B)	Cation at 100% of tetrahedral void
(R)	CsCl type	(C)	Cation at cubic void
(S)	Na_2O type	(D)	Cation at 100% octahedral void

(A) P $\longrightarrow$ 4;Q $\longrightarrow$ 1;R $\longrightarrow$ 3;S $\longrightarrow$ 2

(B) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 4;S $\longrightarrow$ 1

(C) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(D) P $\longrightarrow$ 4;Q $\longrightarrow$ 3;R $\longrightarrow$ 4;S $\longrightarrow$ 2

SECTION-II Second Decimal : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

Rankers Academy JEE

1. A metal 'M' having atomic mass 31.25 crystallizes in cubic close packing and it shows " Schottky defects". If the edge length of the cubic lattice is 500 pm and density of the metal is 1.6075 gm/ml then calculate number of moles of metal atom 'M' missing per litre of the crystal.
[Given : 1 amu = 1.67×10^{-24} gm]

2. If truncated octahedron has x corners y square faces & z hexagon then
z - x/y = ?

3. If NaCl is dopped with 10^{-3} mol % $SrCl_2$ and the numbers of cation vacancies per mole of NaCl is $x \times 10^{18}$, the value of 'x' is ($N_A = 6 \times 10^{23}$)

SECTION-I (i) Single Correct (Maximum Marks:30)

Each question of this section have (+3, 0) marks.

1. The desorption of gas molecules from the adsorbent surface obeys Arrhenius equation. The average time upto which a N_2 molecule may remain adsorbed at Pt-surface at 400K is
[Given: Pre-exponential factor, $A = 1.25 \times 10^8\ s^{-1}$;]
Activation energy of desorption = 16Kcal' $e^{20} = 5 \times 10^8$]

(A) 0.25 sec (B) 4 sec

(C) 8 sec (D) 0.125 sec.

2. Smoke precipitator works on the principle of

(A) distribution law (B) neutralization of charge on colloids

(C) Le-Chaterlier's principle (D) addition of electrolytes

3. When $H_{2(gas)}$ gets adsorbed on transition metals hydride formation takes place. Which of the options will not be correct on the basis of above observation.

(A) Greater the surface area of transition metals greater will be the extent of adsorption.

(B) The enthalpy change involved will be in the range of 20–40 kJ/mole.

(C) The extent of adsorption will increase with increase in temperature.

(D) The adsorption will be irreversible in nature.

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4. Which of the following reaction can result in formation of a colloidal solution?

(A) Reduction of $AuCl_3$ with HCHO in presence of water.

(B) Reduction of $CO_{2(g)}$ using Mg to give carbon & MgO.

(C) Acid catalysed hydrolysis of ester.

(D) oxidation of $NH_{3(g)}$ into nitric oxide in Ostwald's process.

5. Which of the following is not true with respect to catalyst and catalytic action?

(A) A catalyst increases the rate of a reaction while inhibitor decreases the reaction rate.

(B) A promoter cannot catalyse the reaction on its own without the presence of catalyst.

(C) The activity of enzyme catalysts increases in presence of metal ions like Mn^{+2}, Co^{2+} etc.

(D) When a solid catalyst is used for gaseous reaction, energy released in physisorption increases the rate of reaction.

6. Which of the following process does not occur at the interface of phases?

(A) crystallisation (B) heterogenous catalysis

(C) homogeneous catalysis (D) corrosion

7. Which ion has least coagulation value for +ve sol ?

(A) $[Fe(CN)_6]^{4-}$ (B) Cl^- (C) SO_4^{2-} (D) PO_4^{3-}

8. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25°C. For this process, the correct statement is -

(A) The adsorption requires activation at 25ºC

(B) The adsorption is accompanied by a decrease in enthalpy

(C) The adsorption increases with increase of temperature

(D) The adsorption is irreversible

9. A soap ($C_{17}H_{35}COONa$) solution becomes a colloidal sol at a concentration of 1.2×10^{-3} M. On the average, 2.4×10^{13} micelle are present in 1 mm^3. What is the average number of stearate ions in one colloidal particle (micelle) : **Take : Avogadro's number = 6×10^{23}** (Assuming all the ions are present in form of micelle)

(A) 30 (B) 40 (C) 50 (D) 60

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10. Promoters and Poison are the substance use in chemical reaction which

(A) Enhance and decrease the activity of catalyst respectively

(B) Decrease and enhance the activity of catalyst respectively

(C) Do not have any effect on catalyst

(D) Can be used in place of catalyst whenever required

SECTION-I (ii) Multiple Correct : (Maximum Marks: 44)

Each question of this section have (+4, 0) marks.

11. Which of the following are true regarding micelles

(A) Micelles formation leads to net increase in entropy

(B) Hydrophobic portions are oriented within the cluster & the hydrophilic portions are exposed to the water phase

(C) After CMC no change in surface tension takes place, as the surface, becomes completely loaded with surfactant molecules and any further addition must arrange only as micelles.

(D) Micelles are not static, they are dynamic structures, so that the surfactant molecules leave the micelles & go into the solution, while other surfactant molecules enter the micelle from the solution.

12. Colloidal solution can be purified by

(A) Dialysis
(B) Electrodialysis
(C) Electrophoresis
(D) Ultrafiltration

13. Which of the following is not lyophillic

(A) Gelatin sol
(B) Silver sol
(C) Sulphur sol
(D) As_2S_3 sol

14. Colloidal Gold can be prepared by

(A) Bredig's are method
(B) Reduction of $AuCl_3$
(C) Hydrolysis
(D) Peptization

15. The correct statement(s) about surface properties is (are)

(A) Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium
(B) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.
(C) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution.
(D) The critical temperatures of ethane and nitrogen and 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature.

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16. Choose the correct reason(s) for the stability of the lyophobic colloidal particle.

(A) Preferential adsorption of ions on their surface from the solution
(B) Preferential adsorption of solvent on their surface from the solution
(C) Attraction between different particles having opposite charges on their surface
(D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles

17. Tyndal effect is appriciable when

(A) The diameter of dispersed phase is comparable to the wavelength of light used.
(B) The diameter of dispersed phase is much smaller than the wavelength of light used
(C) The refractive indices of the dispersed phase and the dispersion medium must be same
(D) The refractive indices of the dispersed phase and the dispersion medium must differ greatly in magnitude.

18. Which of the following is/are true regarding given colloids :

(A) In sol solids dispersed in liquids

(B) In gels liquids dispersed in solids

(C) In emulsion liquid dispersed in liquids

(D) Smoke is an aerosol in which gas dispersed in solid

19. Which of the following is/are correct statement :

(A) Milk is an oil in water type emulsion

(B) Micelle formation takes place above a temperature called kraft temperature (T_k)

(C) Emulsifying agent are the component used to stabilize emulsion

(D) Protein and heavy metal salts are common emulsfying agent for O/W and W/O emulsions respectively

20. Which of the following statements are true about AgI sol prepared by peptization of AgI with $AgNO_3$.

(A) In electrophoresis, the dispersed phase will move towards cathode.

(B) $AlCl_3$ is more effective than Na_3PO_4 for bringing about coagulation of sol.

(C) Na_3PO_4 is more effective than $AlCl_3$ for bringing about coagulation of sol.

(D) On persistant dialysis of sol, precipitation of coloidal sol take place (coagulation)

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21. Sodium palmitate is a major constituent in soap. When an aqueous solution of sodium palmitate having in 1 kg of solvent is analysed for its boiling point, following data is obtained.

Moles	0	0.02	0.04	0.05	0.06	0.08
Boiling point	100°C	100.0372°C	100.0744°C	100.093°C	100.1°C	100.12°C

Identify the option(s) which is/are correct.

(A) Aqueous mixture of sodium palmitate having concentration equal to 0.07M will be heterogenous.

(B) The CMC value of sodium palmitate should be less than 0.06M.

(C) Osmotic pressure at 0.08 M at 300 K will be 1.9704 atm.

(D) At 0.4M concentration, the solution is homogenous.

SECTION-I (iii) Paragraph : (Maximum Marks: 6)

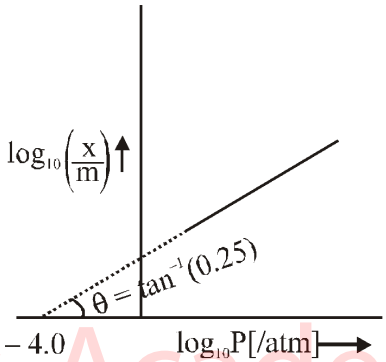
Each question of this section have (+3, 0) marks.

Paragraph for Questions 22 and 23

The adsorption of a gas at a metal surface is called occlusion. The extent of gas adsorbed at metal surface at a certain temperature depends on pressure as

$$\frac{x}{m} = K.P^{\frac{1}{n}}$$

Where 'K' and 'n' are constant for a particular combination of gas and metal at fixed temperature and $\frac{x}{m}$ is the mass of gas adsorbed per unit mass of metal. For H_2 gas adsorbing at Pt-surface, the following graph is obtained.



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22. The correct values of 'K' and 'n' are respectively.

(A) 1,0.25 (B) 10, 4 (C) 10, 0.25 (D) 0.10, 4.0

23. The mass of H_2 gas adsorbed on the surface of 10 gm Pt at 16 atm is

(A) 20 gm (B) 2 gm (C) 4 gm (D) 200 gm

SECTION-II Second Decimal : (Maximum Marks: 15)

Each question of this section have (+3, 0) marks.

1. Gold number of Haemoglobin is 0.030. Hence 100 ml of gold sol will require haemoglobin (in mg) so that gold is not coagulated by 10 ml of 10% NaCl solution.

2. A colloid prepared by the addition of KI to $AgNO_3$ solution is purified using dialysis. Find the minimum mass in grams of an electrolyte 'AB' (GMM = 60) required to completely coagulate 1 ℓ of the aforementioned colloid

Given :

Active ion causing flocculation	Flocculation value(mmole/ℓ)
A^+	50
B^-	100

3. How many of the following statements are CORRECT -

(i) During adsorption, there is always a decrease in residual forces of the surface

(ii) During adsorption, there is decrease in surface energy which appears as heat.

(iii) Physisorption can be multilayered.

(iv) The ppt of $Mg(OH)_2$ attain blue color in presence of magneson, due to adsorption of magneson

(v) A mixture of noble gases can be separated by adsorption on coconut charcoal at different temperature.

4. How many of the following phenomenon are observed due to process of coagulation.

(A) Delta formation at the meeting point of river and sea.

(B) Blue colour of sky.

(C) Precipitation of coal ash from smoke in chimney's.

(D) Destruction of lyophobic solution.

(E) Brownian motion.

(F) Cleaning action of detergents.

(G) Artificial rain.

(H) Use of alums in cleaning water.

5. Calculate the number of sols which are negatively charged.

$Fe_2O_3.xH_2O$ sol Sb_2S_3 sol

Ag sol Cu sol

Blood basic dye

sol of clay sol of starch

sol of sodium stearate sol of sodium laurylsulphate

sol of charcoal.

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SECTION-I (i) Single Correct (Maximum Marks:27)

Each question of this section have (+3, 0) marks.

1. The wavelength of K_α-characteristic X-rays produced is λ, when cathode rays strike a metal of atomic number Z. What should be the atomic number of metal such that it can produce the K_α-characteristic X-rays of wavelength 4λ?

(A) $\frac{Z}{16}$ (B) $\frac{Z}{2}$

(C) $\frac{Z+1}{2}$ (D) $2Z - 1$

2. According to Maxwell's theory of electrodynamics, an electron going in a circle should emit radiation of frequency equal to its frequency of revolution. What should be the wavelength of the radiation emitted by a hydrogen atom in the ground state if this rule is followed?

(A) 4500 nm (B) 450 nm

(C) 45 nm (D) 4.5 nm

3. An electron revolving around H-nucleus in ground state absorbs 10.2 eV energy. Its angular momentum increases by

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(A) $\frac{h}{2\pi}$ (B) $\frac{h}{\pi}$

(C) $\frac{2h}{\pi}$ (D) $\frac{h}{4\pi}$

4. In a hydrogen like species , an electron jumps from an orbital having two radial and two angular nodes to the orbital having same sign of wave function in all direction, at any distance. If the energy of emitted photon is 326.4 eV, the atom or ion is

(A) H (B) Li^{2+}

(C) He^{+} (D) B^{4+}

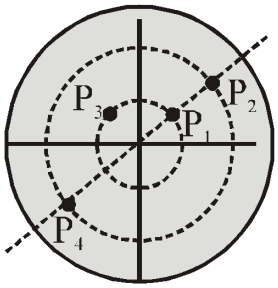
5. O_2 undergoes photochemical dissociation into one normal oxygen atom and one oxygen atom, 1.2 eV more energetic than normal. The dissociation of O_2 into two normal atoms of oxygen requires 482.5 kJ/mol. The maximum wavelength effective for photochemical dissociation of O_2 is

(1 eV = 96.5 kJ/mol)

(A) 248 nm (B) 1033.3 nm

(C) 1236.2 nm (D) 200 nm

6. The given diagram shows points P_1, P_2 P_3 & P_4 in 1s orbital.



1s orbital

The correct order of radial probability density at these point are

(A) $P_1 = P_2 = P_4 > P_3$ (B) $P_1 = P_3 < P_2 = P_4$

(C) $P_1 = P_3 > P_2 = P_4$ (D) $P_1 = P_2 = P_4 < P_3$

7. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon

(A) 3s (B) 2p

(C) 2s (D) 1s

8. Assuming Heisenberg Uncertainity Principle to be true what could be the minimum uncertainty in de-broglie wavelength of a moving electron accelerated by Potential Difference of 6 V whose uncertainty in position is $\frac{7}{22}$ n.m.

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(A) 6.25 Å (B) 6 Å

(C) 0.625 Å (D) 0.3125 Å

9. A light source of wavelength λ illuminates a metal and ejects photo-electrons with $(KE)_{max}$ = 1 eV. Another light source of wavelength, $\frac{\lambda}{3}$, ejects photo-electrons from same metal with $(KE)_{max}$ = 4 eV. Find the value of work function ?

(A) 1 eV (B) 2 eV

(C) 0.5 eV (D) None of these

SECTION-I (ii) Multiple Correct : (Maximum Marks: 20)

Each question of this section have (+4, 0) marks.

10. The orbital angular momentum of an electron in a single electron system is $\sqrt{3}\frac{h}{\pi}$. Which of the following angular momentum value (s) are not possible for this electron in Bohr orbit.

(A) $\frac{3h}{2\pi}$ (B) $\frac{h}{\pi}$

(C) $\frac{3h}{\pi}$ (D) $\frac{2h}{\pi}$

11. For He^+ ion, select incorrect option(s) :

(A) Difference between radius of II & III Bohr's orbit is greater than that of III & IV Bohr orbit.

(B) Ratio of velocity of electron in III & IV Bohr's orbit is less then one.

(C) Energy difference between consecutive energy levels decreases continuously.

(D) Ratio of wavelength of least energetic and most energetic photon of lyman series is (3/4).

12. Which of the following orbitals has (have) two spherical nodes?

(A) 2s (B) 4p

(C) 3d (D) 6f

13. Orbitals in which the numbers of radial and angular nodes are equals?

(A) 3p (B) 4d (C) 5d (D) 6f

14. When photons of energy 4.25 eV strike the surface of a metal 'A', the ejected photo electrons if maximum kinetic energy have wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal 'B' by photons of energy 4.7 eV is 1.5 eV less than that of photo electrons of metal A. If wavelength of photo electrons of metal B is twice than that of photoelectron of 'A' then -

(A) Work function of A = 2.25 eV

(B) Work function of B = 4.2 eV

(C) Maximum (K.E.) of photoelectrons of 'A' = 2 eV

(D) Maximum (K.E.) of photoelectrons of 'B' = 2.75 eV

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SECTION-I (iii) Paragraph : (Maximum Marks: 18)

Each question of this section have (+3, 0) marks.

Paragraph for Questions 15 - 17

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

15. The state S_1 is :-

(A) 1s (B) 2s (C) 2p (D) 3s

16. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-

(A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

17. The orbital angular momentum quantum number of the state S_2 is :-

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Questions 18 - 20

The location of electron inside the atom can be predicted by wave mechanical model which gives wavefunction of orbitals represented as $\Psi_{n,\ell,m}$ where n, ℓ, m represents quantum numbers sometimes, Ψ can be analysed as two separate components, radial and angular, represented as $\Psi_{(r)}$ and $\Psi_{(\theta,\phi)}$ respectively for H, the following expressions of $\Psi_{n,\ell,m}$ are known

[Where a_0 is constant and σ is $\frac{2r}{na_0}$]

$$\Psi_{1,0,0} = \frac{2}{a_0^{3/2}} e^{-\sigma/2} \qquad \Psi_{3,0,0} = \frac{1}{9\sqrt{3}} \left(\frac{1}{a_0}\right)^{3/2} (6 - 6\sigma + \sigma^2)\, e^{-\sigma/2}$$

$$\Psi_{2,0,0} = \frac{1}{2\sqrt{2}} \left(\frac{1}{a_0}\right)^{3/2} (2 - \sigma)\, e^{-\sigma/2} \qquad \Psi_{3,1,0} = \frac{1}{9\sqrt{3}} \left(\frac{1}{a_0}\right)^{3/2} (4 - \sigma)\, \sigma e^{-\sigma/2}$$

18. For 1 s orbital H-atom, the radial probability will be maximum at what radial distance from nucleus:

(A) r = 0 (at the nucleus)

(B) r = ∞

(C) r = a_0

(D) r = $2a_0$

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19. At what radial distance will there be a spherical node for $3p_y$ orbital of hydrogen?

(A) r = $6a_0$

(B) r = $2a_0$

(C) r = 0

(D) data is insufficient.

20. Which of the following statements are incorrect w.r.t. $\Psi_{(r)}$ v/s r graph for H-atom

(A) For 1s orbital the graph will not intersect x - axis at finite distance.

(B) For 2s orbital, the graph will have one 'local minima' only.

(C) For 3s orbital, the graph will intersect x - axis at two distinct points.

(D) For 3p orbital, the graph will intersect x-axis at two non-zero distinct points.

SECTION-II Second Decimal : (Maximum Marks: 9)

Each question of this section have (+3, 0) marks.

1. How many non-spherical orbitals are possible that have atleast two maxima if a curve is plotted between radial probability distribution function versus radial distance for which principal quantum number: $n \leq 4$?

2. If 4 times the most probable distance of electron of a 1s orbital in a unielectronic atom /ion is given by 0.529 Å then the atomic number of the unielectronic species will be.

   [**Given :** Wave function of 1s orbital, $\Psi = \sqrt{\left(\frac{Z^3}{\pi a_0^3}\right)} e^{-Zr/a_0}$ where a_0 = radius of first Bohr's orbital in H − atom ≡ (52.9pm) and Z is atomic number of unielectrnic species]

3. Find the number of spectral lines in Paschen series emitted by an atomic hydrogen, when electron is excited from ground state to 7th energy level returns back.

SECTION-IV Matrix Match : (Maximum Marks: 8)

Each question of this section have (+2, 0 each) marks

1. Match the column -

Column-I		Column-II	
(A)	ℓ	(P)	Indicates the distance from nucleus
(B)	s	(Q)	It can have only two possible values.
(C)	m	(R)	Possible values are 0 to (n − 1)
(D)	n	(S)	Indicates the possible orientations in space.
		(T)	Possible values are +ℓ to −ℓ including 'Zero'

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GRP # 1

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	D	D	D	C	B	A	C	B
SECTION-I (ii) Multiple Correct	Q.	9	10	11	12	13			
	A.	A,B,C	A,D	A,C	A,B	C,D			
SECTION-II Second Decimal	Q.	1	2	3	4				
	A.	3.00	3.00	2.94	16.10				
SECTION-IV Matrix Match	Q.	1							
	A.	A->PQRS,B->PQRS,C->PQRST,D->PQ							

GRP # 2

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9
	A.	B	B	B	C	A	C	D	B	A
SECTION-I (ii) Multiple Correct	Q.	10	11	12						
	A.	C	A,B,C	B,D						
SECTION-I (iii) Paragraph	Q.	13	14	15						
	A.	B	C	A						
SECTION-II Second Decimal	Q.	1	2	3	4					
	A.	1.00	3.20	1.00	4.00					

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GRP # 3

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6
	A.	C	B	A	B	D	A
SECTION-I (ii) Multiple Correct	Q.	7	8	9	10	11	
	A.	A,C	B	B,D	B,C,D	A,B	
SECTION-I (iii) Paragraph	Q.	12	13	14			
	A.	C	C	C			
SECTION-II Second Decimal	Q.	1	2	3			
	A.	99.50	2.40	-17.80			
SECTION-IV Matrix Match	Q.	1					
	A.	A->PRST,B->PT,C->PRST,D->Q					

GRP # 4

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9
	A.	C	B	C	B	C	B	B	A	C
SECTION-I (ii) Multiple Correct	Q.	10	11	12						
	A.	B,D	A,C	B,D						
SECTION-I (iii) Paragraph	Q.	13	14	15						
	A.	B	C	D						
SECTION-II Second Decimal	Q.	1	2	3	4					
	A.	100.00	4.00	5.00	6.00					
SECTION-IV Matrix Match	Q.	1								
	A.	A->PT,B->PQRT,C->PQRS,D->PRT								

GRP # 5

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9
	A.	B	B	D	B	A	C	B	D	A
SECTION-I (ii) Multiple Correct	Q.	10	11							
	A.	B,C	A,B,D							
SECTION-I (iii) Paragraph	Q.	12	13	14						
	A.	A	C	B						
SECTION-II Second Decimal	Q.	1	2	3						
	A.	4.00	1.00	5.00						
SECTION-IV Matrix Match	Q.	1								
	A.	A->QR,B->PS,C->QR								

GRP # 6

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6
	A.	C	B	A	C	B	B
SECTION-I (ii) Multiple Correct	Q.	7	8	9	10		
	A.	A,B,D	A,C,D	B,C	A,B,C,D		
SECTION-I (iii) Paragraph	Q.	11	12	13			
	A.	C	A	B			
SECTION-II Second Decimal	Q.	1	2	3	4	5	
	A.	0.19	3.00	4.00	8.00	-0.54	
SECTION-IV Matrix Match	Q.	1					
	A.	A->S,B->R,C->QT,D->P					

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GRP # 7

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	A	D	C	A	D	A	D	C
SECTION-I (ii) Multiple Correct	Q.	9	10	11	12	13			
	A.	A,B,C	A,B,C	A,C	A,B,D	A,D			
SECTION-I (iii) Paragraph	Q.	14	15	16	17	18			
	A.	B	B	B	D	B			
SECTION-II Second Decimal	Q.	1	2	3					
	A.	8.00	7.00	4.00					
SECTION-IV Matrix Match	Q.	1							
	A.	A->PRS,B->PRS,C->QS,D->PQS							

GRP # 8

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9
	A.	C	A	A	D	B	C	A	A	A
SECTION-I (ii) Multiple Correct	Q.	10	11	12	13					
	A.	A,C,D	B,C,D	C	A,B,D					
SECTION-I (iii) Paragraph	Q.	14	15	16	17	18	19			
	A.	A	A	B	A	C	B			
SECTION-II Second Decimal	Q.	1	2	3	4					
	A.	200.00	2.00	4.00	4.00					

GRP # 9

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	D	A	B	D	A	B	D	B
SECTION-I (ii) Multiple Correct	Q.	9	10	11	12				
	A.	D	B,C	A,C,D	A,B,C				
SECTION-I (iii) Paragraph	Q.	13	14	15					
	A.	C	B	A					
SECTION-II Second Decimal	Q.	1	2	3					
	A.	1.00	8.00	4.00					
SECTION-IV Matrix Match	Q.	1							
	A.	A->PS,B->R,C->PQ,D->R							

GRP # 10

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8
	A.	C	A	B	C	C	C	D	A
SECTION-I (ii) Multiple Correct	Q.	9	10	11	12				
	A.	C	A,D	A,C,D	B,C				
SECTION-I (iii) Paragraph	Q.	13	14	15	16	17			
	A.	B	A	D	B	D			
SECTION-I (iv) Matching List	Q.	18							
	A.	A							
SECTION-II Second Decimal	Q.	1	2	3					
	A.	2.00	4.00	6.00					

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GRP # 11

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	A	D	C	A	B	A	A
SECTION-I (ii) Multiple Correct	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A,B,C,D	A,B,D	B,C,D	A,B	B,D	A,D	A,D	A,B,C	A,B,C,D	A,C,D
	Q.	21									
	A.	A,B									
SECTION-I (iii) Paragraph	Q.	22	23								
	A.	B	D								
SECTION-II Second Decimal	Q.	1	2	3	4	5					
	A.	0.30	6.00	5.00	5.00	9.00					

GRP # 12

SECTION-I (i) Single Correct	Q.	1	2	3	4	5	6	7	8	9
	A.	C	C	A	D	D	C	D	C	C
SECTION-I (ii) Multiple Correct	Q.	10	11	12	13	14				
	A.	A,B	A,B,D	B,D	A,C	A,B,C				
SECTION-I (iii) Paragraph	Q.	15	16	17	18	19	20			
	A.	B	C	B	C	A	D			
SECTION-II Second Decimal	Q.	1	2	3						
	A.	11.00	4.00	4.00						
SECTION-IV Matrix Match	Q.	1								
	A.	A->R,B->Q,C->ST,D->P								